

Biochip Course

Microfluidic Components

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A Library of Fluidic Building Blocks

1. Channels
2. Valves
3. Mixers
4. Pumps
5. Fluidic connectors

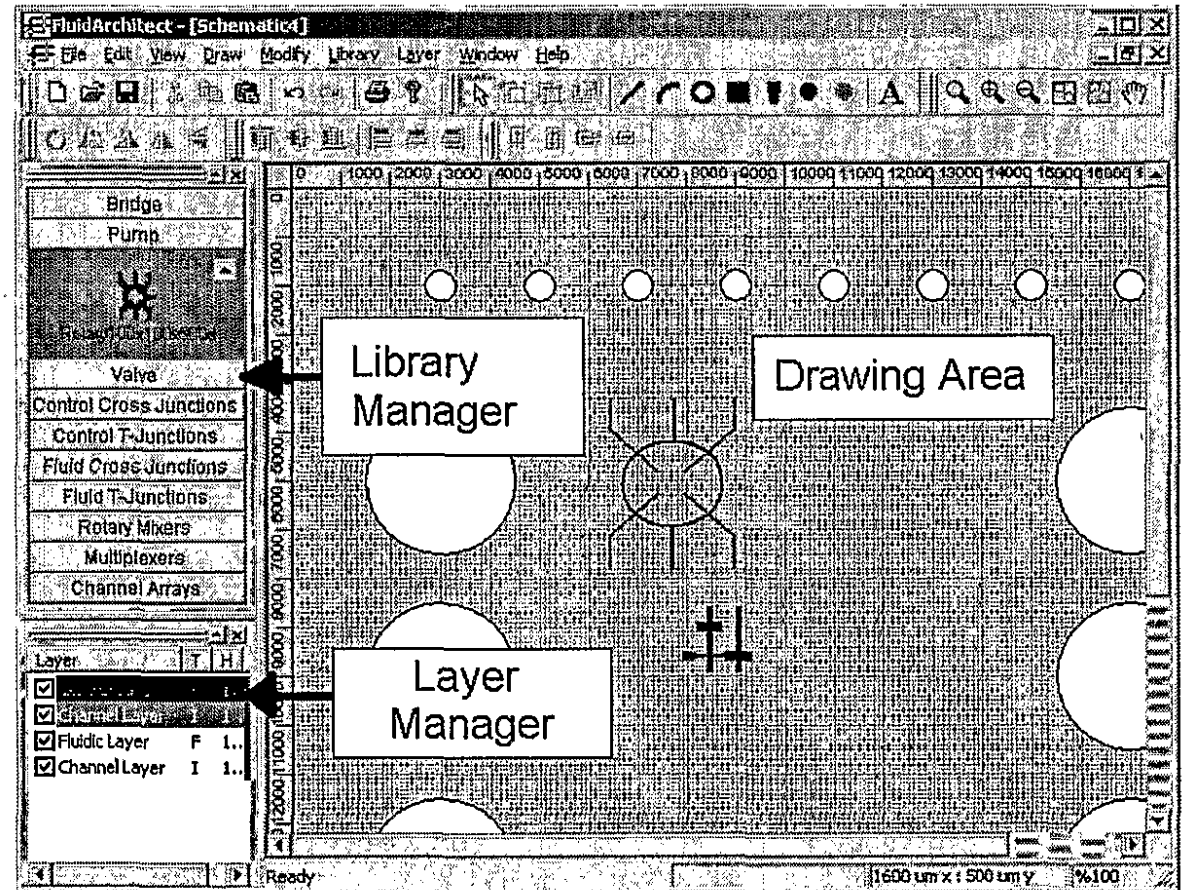


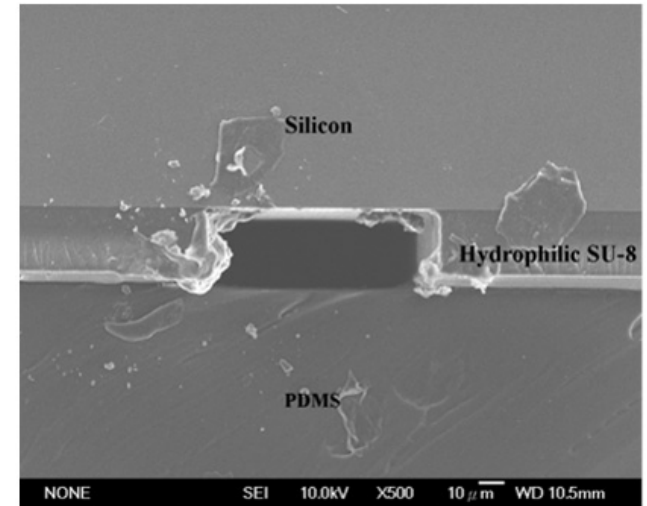
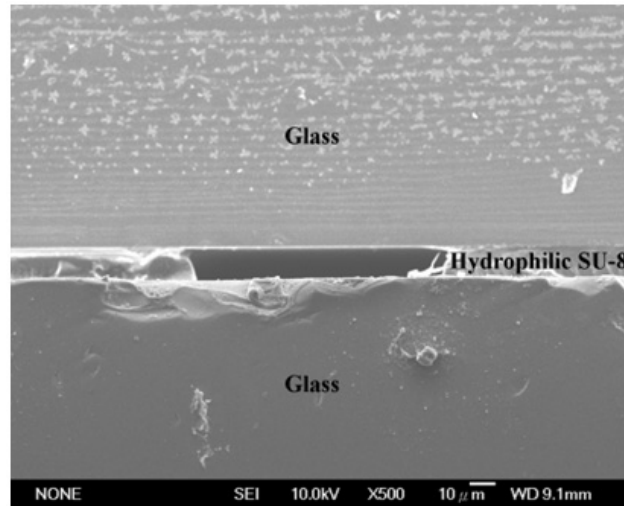
Figure 60 – FluidArchitect's User Interface

<http://www.google.com/patents/WO2002000343A2?cl=en>



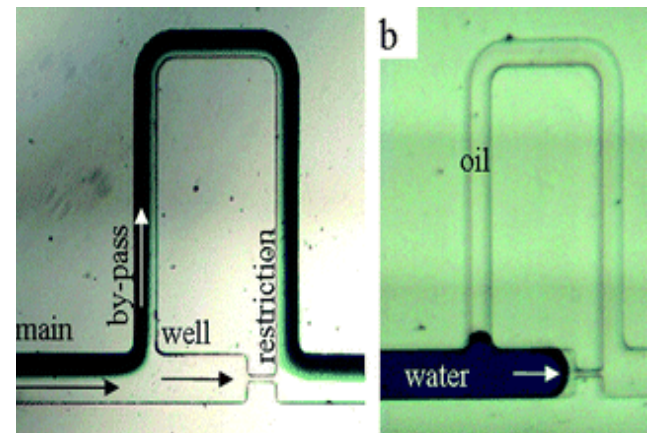
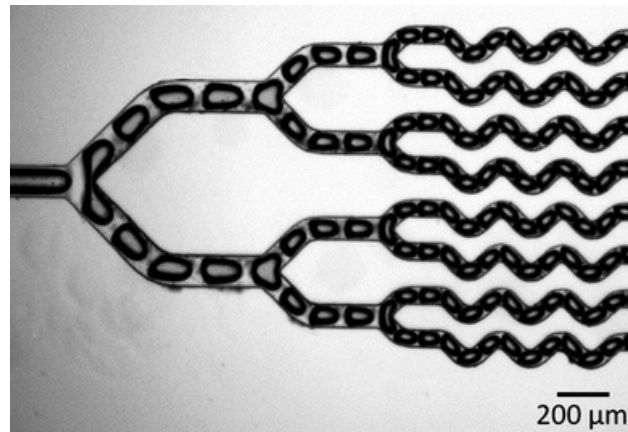
Channels:

Cross section



Bifurcations:

Top view



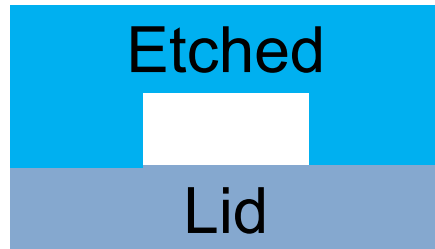
Reservoirs, reactors, detection areas (chambers)

Design choices: materials, size – *Computational Fluid Dynamics*



Channels Cross Sections

Typically fabricated with planar processes:



- Silicon
- Glass
- Plastic
- Elastomer (PDMS)

Adhesive Interposer

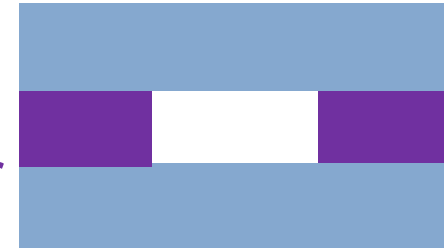
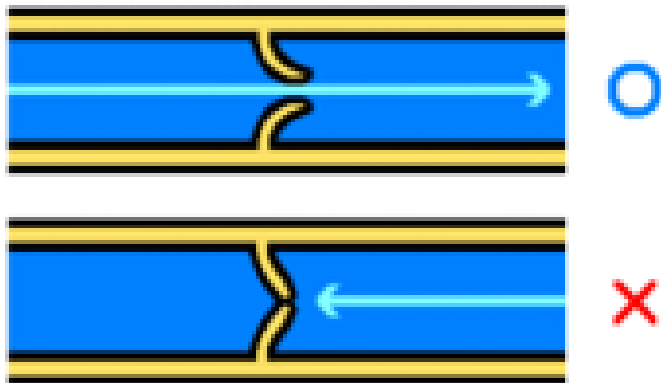
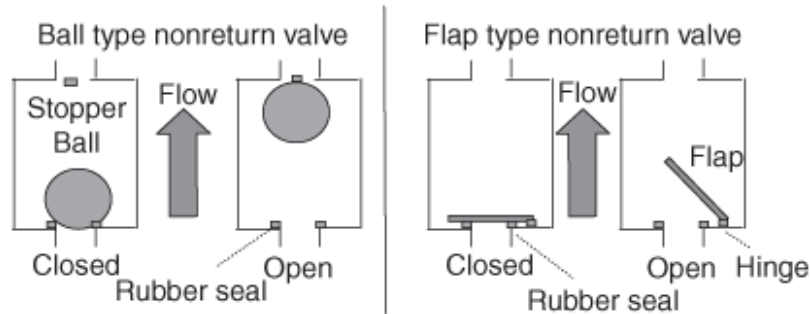


Photo-patternable and adhesive layer

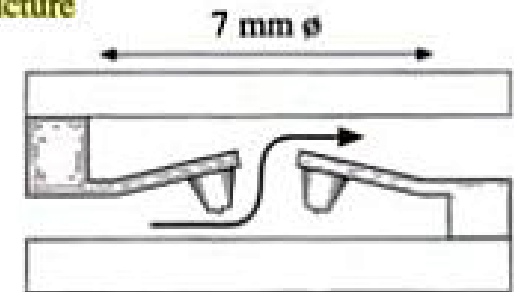
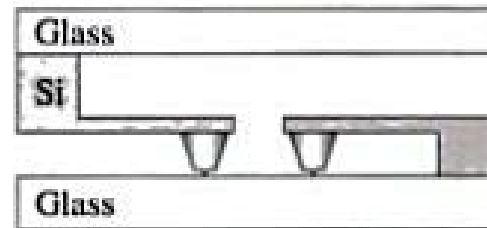
- Wet Resist (SU8)
- Dry Resist
- Adhesives

Passive Monodirectional Valves

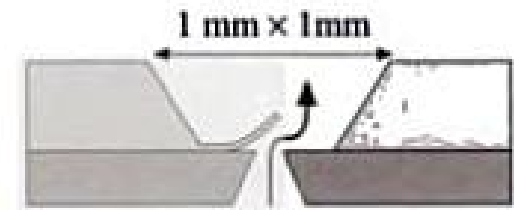
Based on pressure differences and asymmetrical flexible structures



a) Membrane type with MESA structure



b) Cantilever or flap valve



c) Poly-silicon membrane



d) V-groove type





Tesla Valve (Fluidic Diode)

Patented by Nikola Tesla in 1920, it is a one-directional fluidic conduit (due to different fluidic resistances, up to 200:1). No moving parts, but leakage.

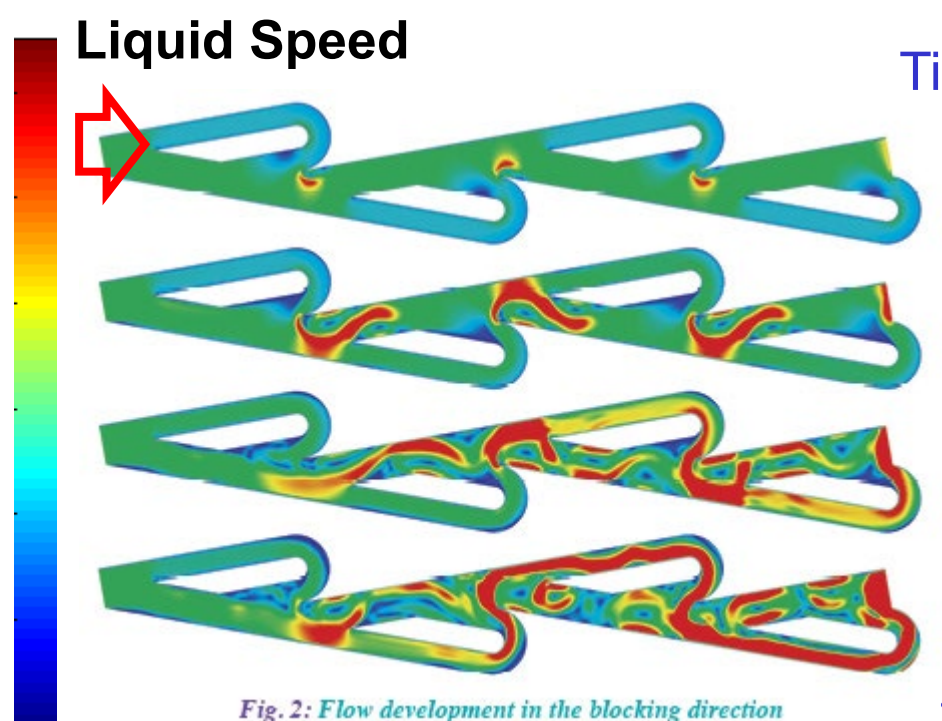
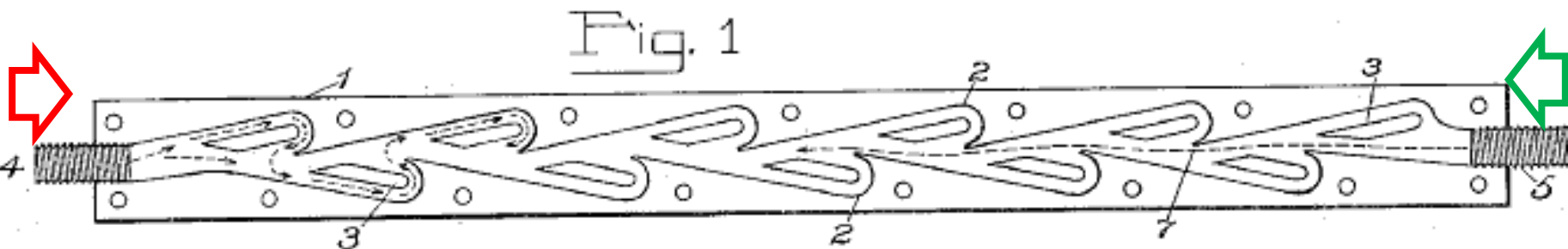


Fig. 2: Flow development in the blocking direction

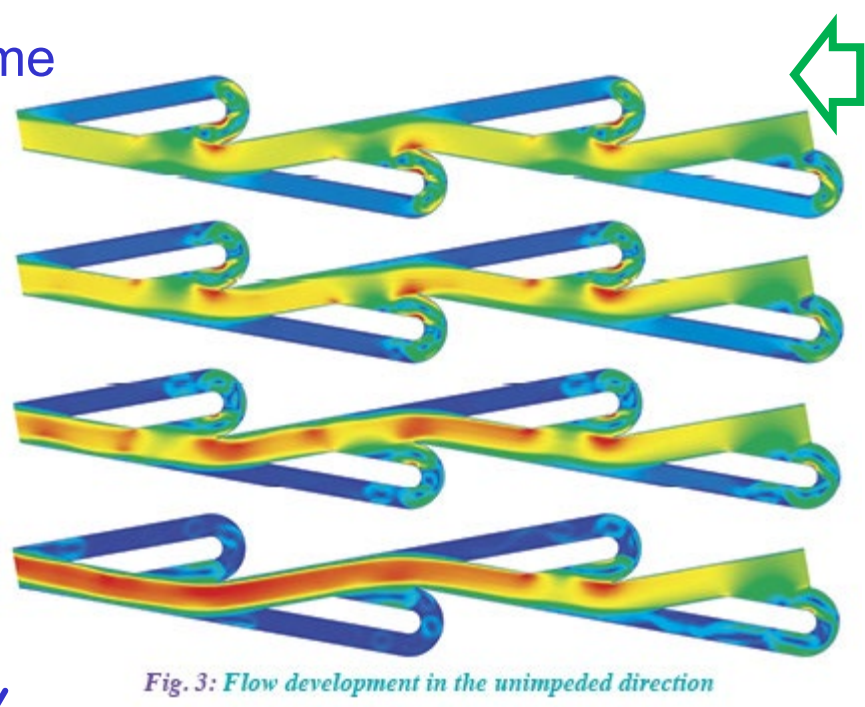


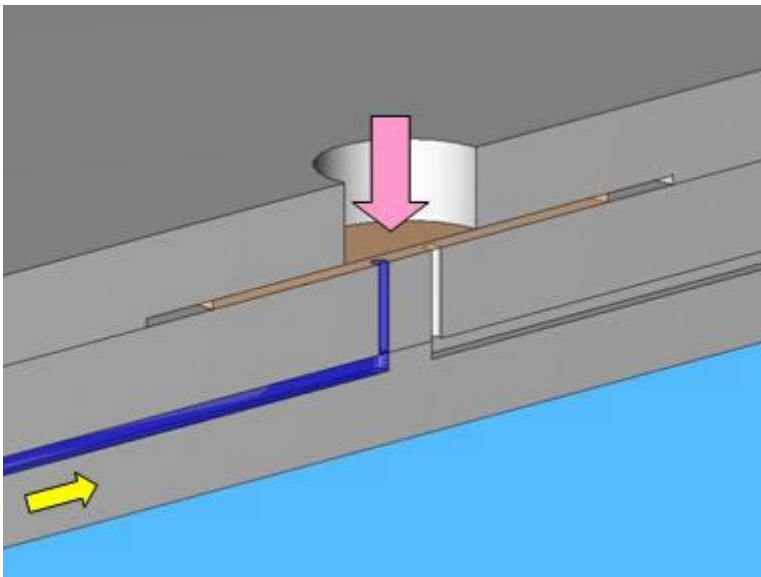
Fig. 3: Flow development in the unimpeded direction



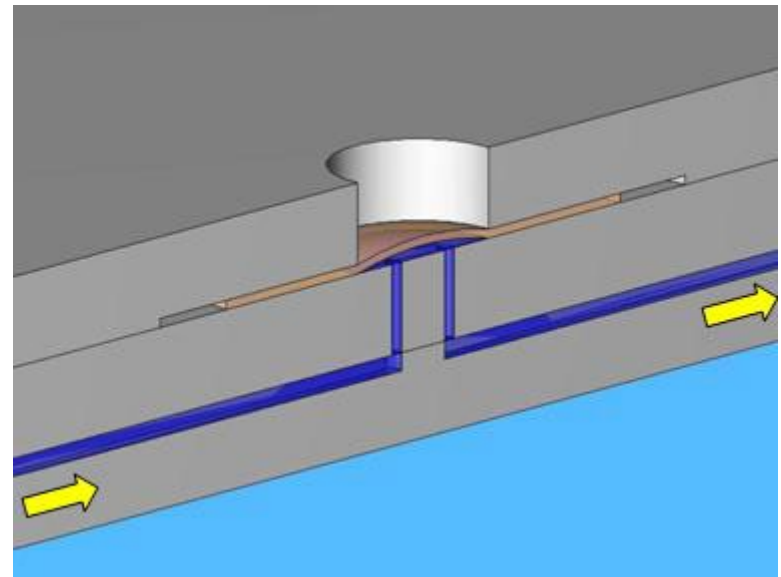
Active Valves

A mechanical actuator controls the valve membrane:

Closed



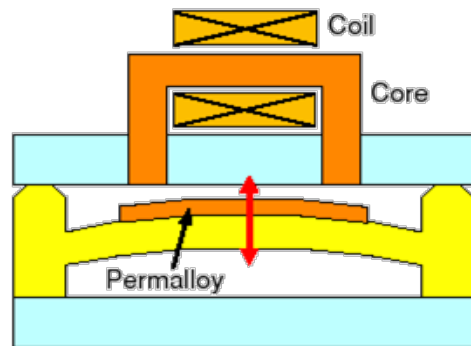
Open



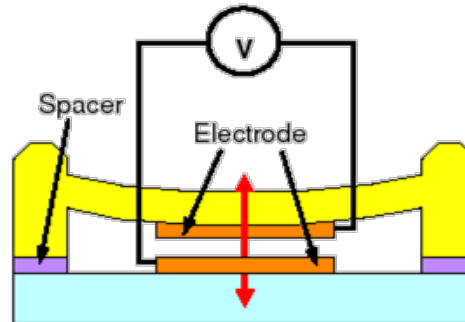
Better performances than passive valves (lower leakage, faster response) but dissipate power.



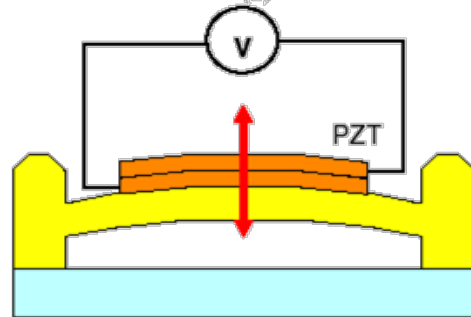
Actuation Types



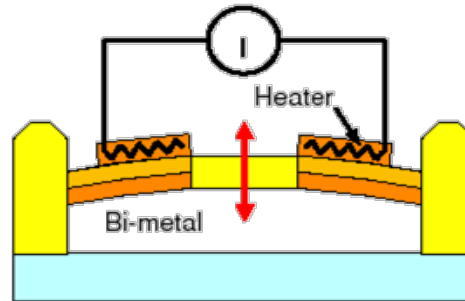
(a)



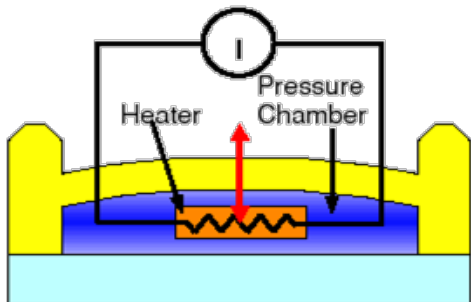
(b)



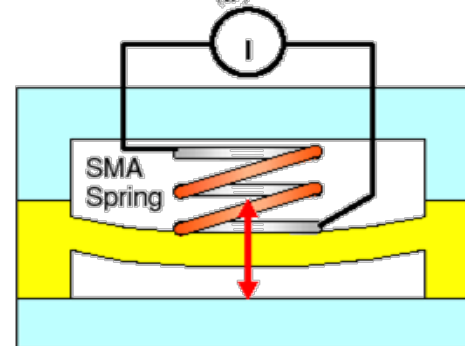
(c)



(d)



(e)



(f)

Electrical drive, mechanical displacement of the membrane

Actuators types:

- (a) Magnetic
- (b) Electrostatic
- (c) Piezoelectric
- (d) Thermal (bimetallic strip)
- (e) Thermal (expansion)
- (f) Shape Memory Alloy



Electrostatic Force

Electrostatic force:

$$F_{ES} = \frac{1}{2} \frac{\partial C}{\partial z} V^2$$

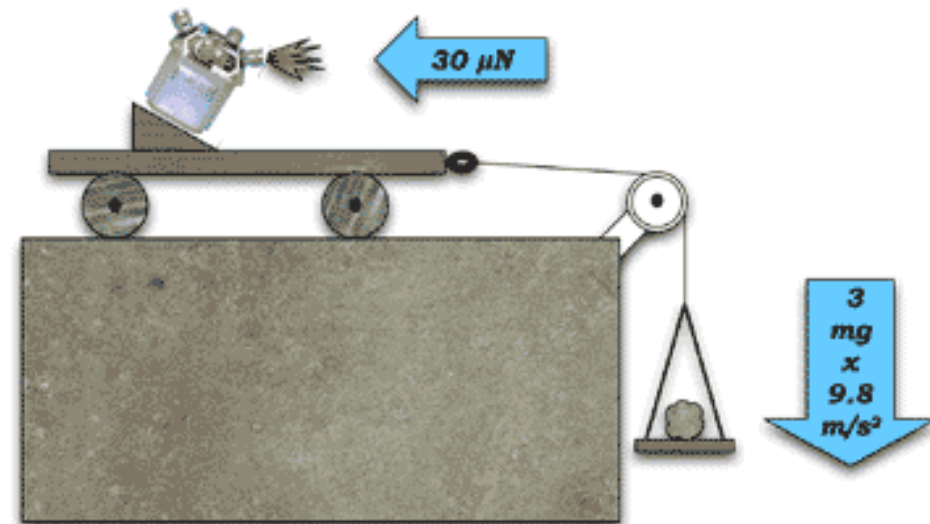
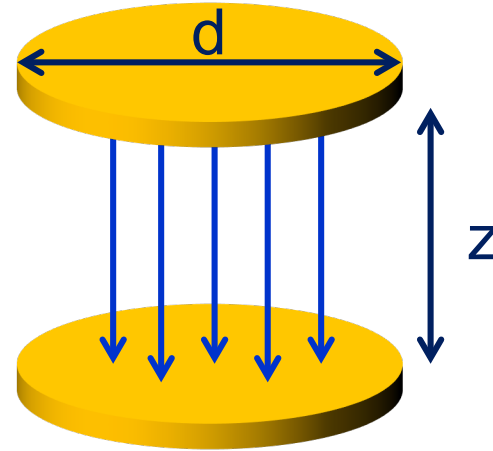
$$C = \varepsilon \frac{A}{z} = \varepsilon \frac{\pi d^2}{4z}$$

$$F_{ES} = -\frac{\varepsilon \pi d^2}{8z^2} V^2$$

If $d = 2\text{cm}$ and $z = 100\mu\text{m}$, and $V = 10\text{V}$

$$F_{ES} = 3.5\mu\text{N}$$

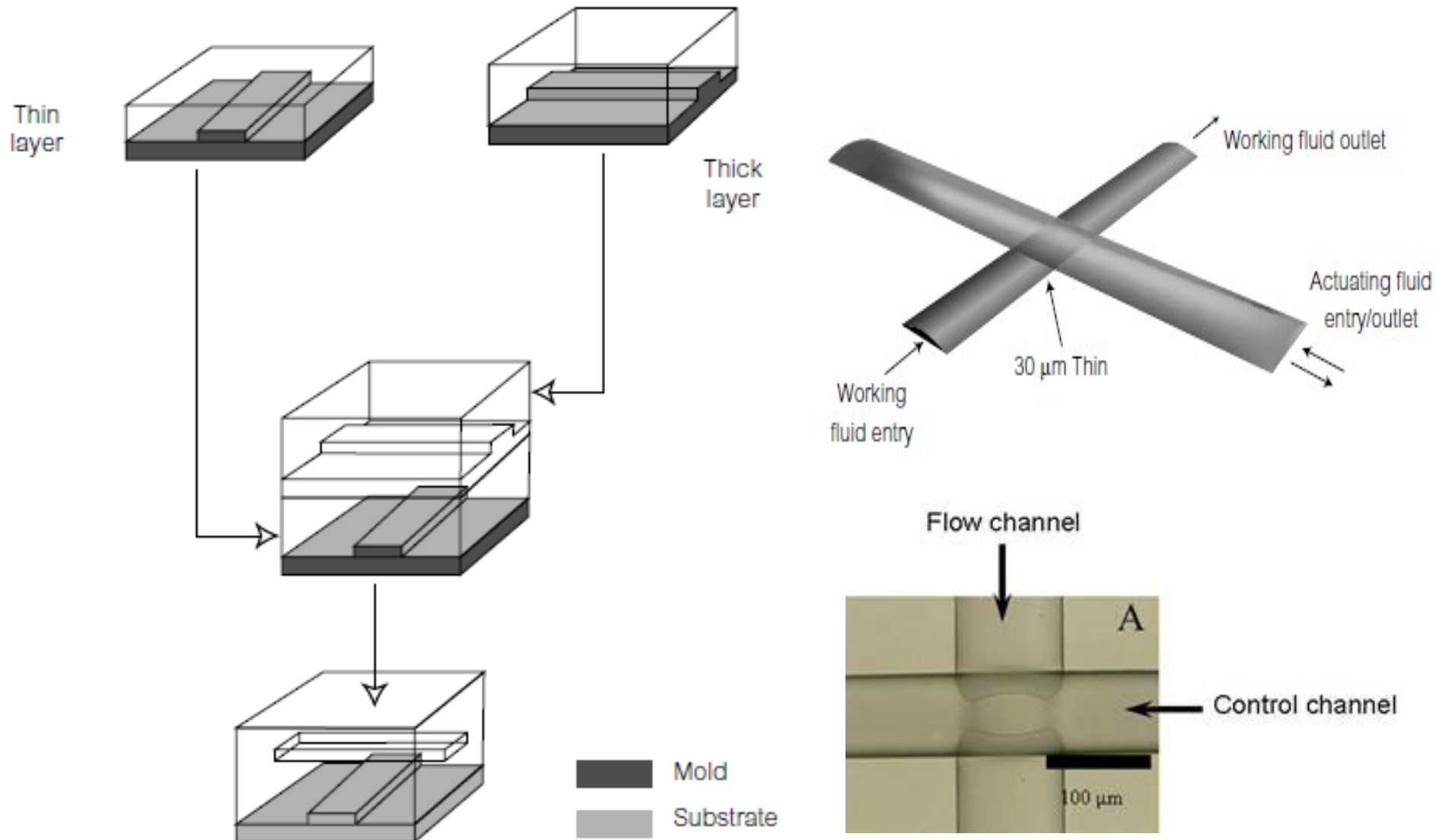
(covalent bond breaks $1.6\mu\text{N}$)





Membrane Microvalves: Hydraulic Control

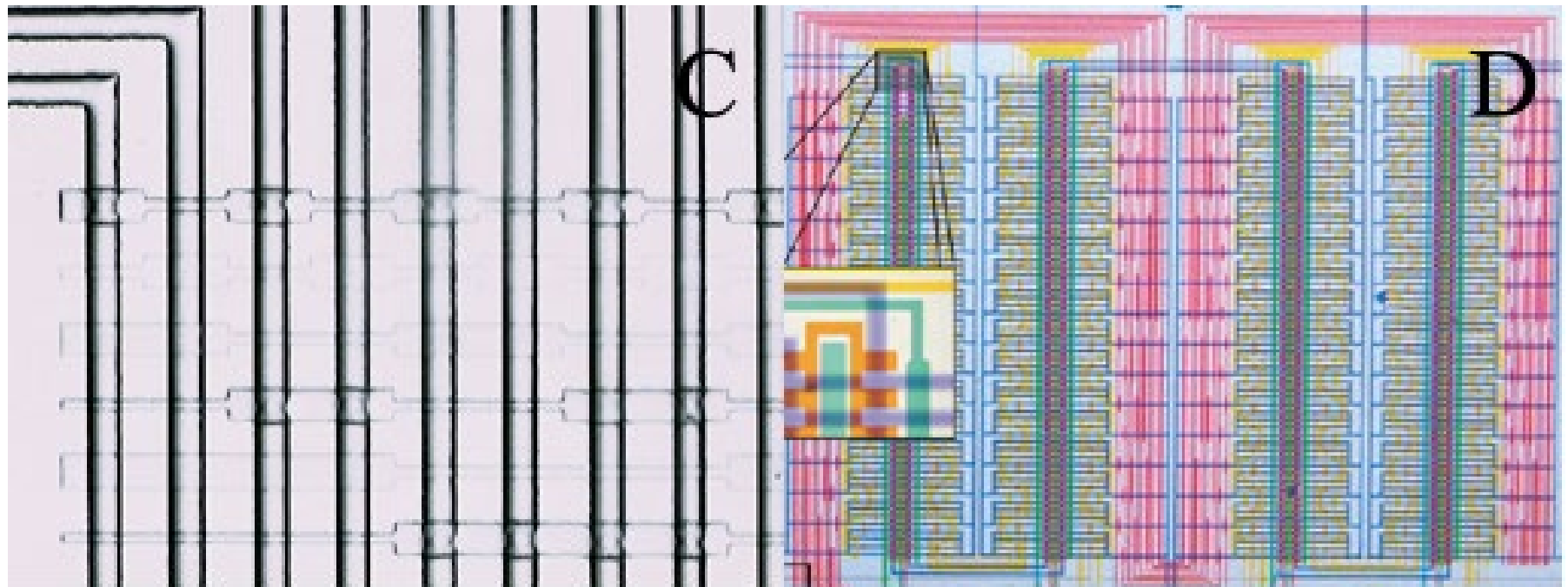
<http://www.stanford.edu/group/foundry/Microfluidic%20valve%20technology.html>





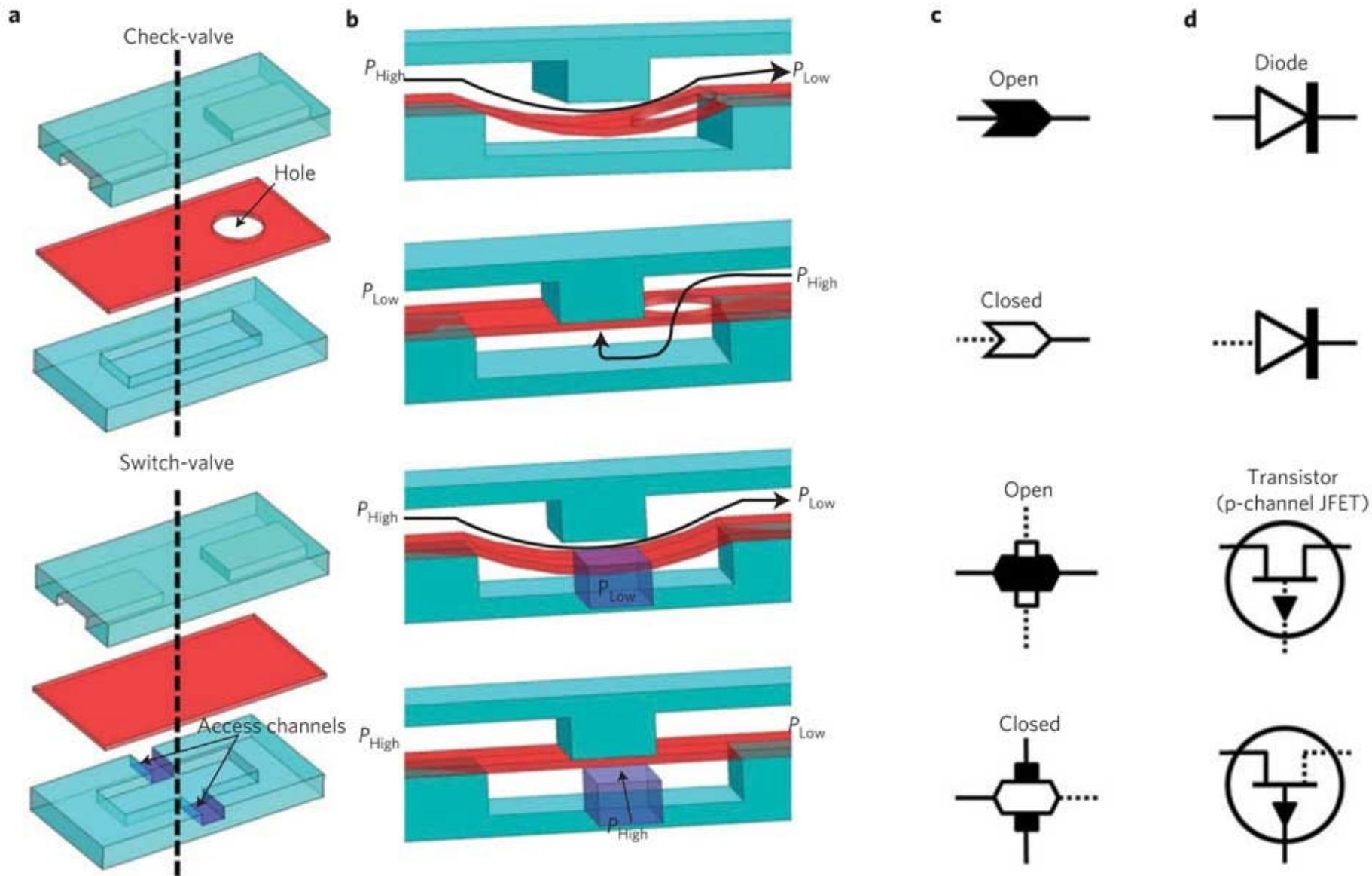
Large Scale of Integration

Demonstrated a density up to 1000 valves per cm^2



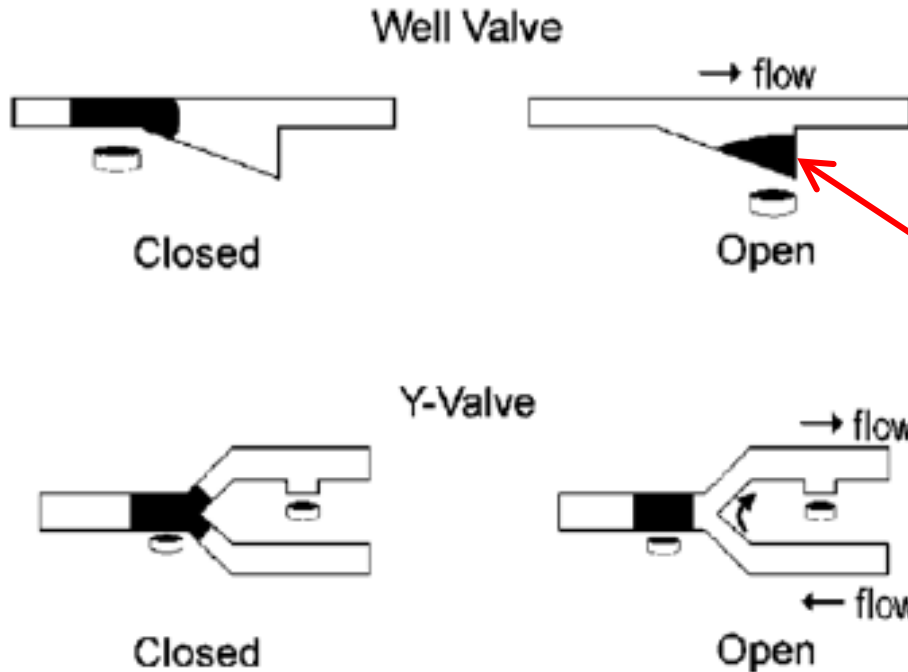


Multilayer Technology



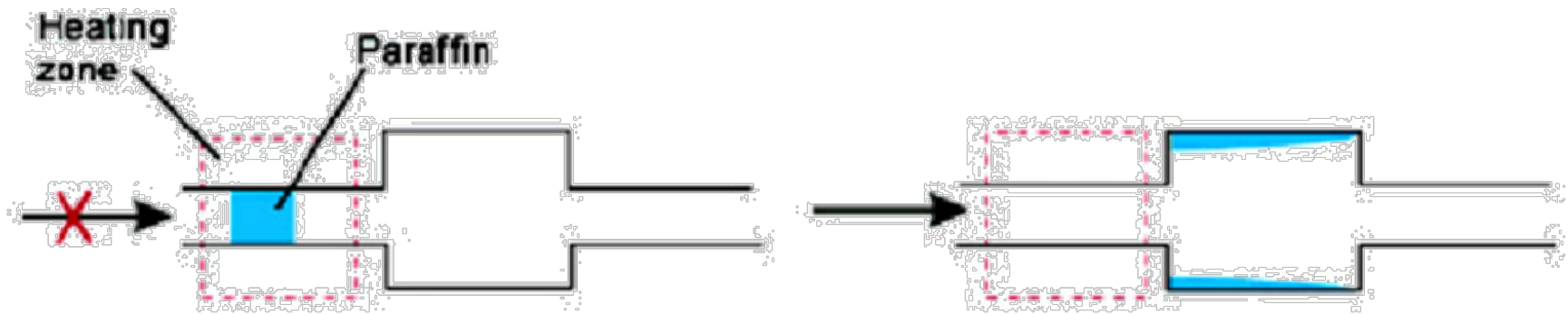


Other Approaches



Magnetically controlled
immiscible fluid
(oil droplet with iron filings)

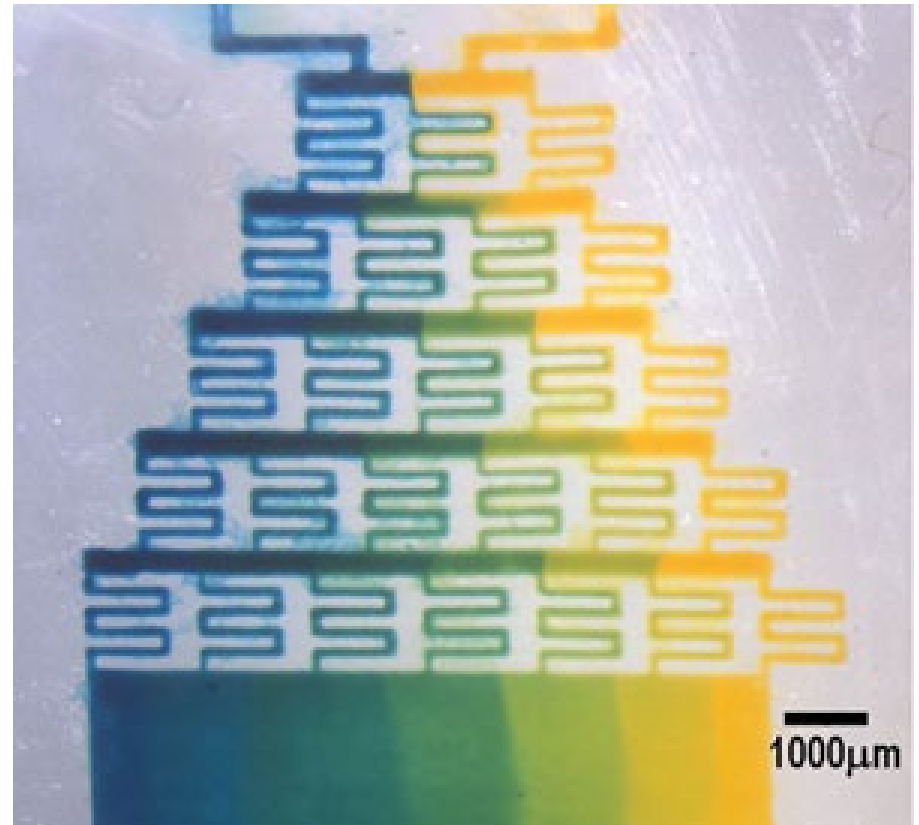
Irreversible valves:





Mixing is a fundamental task to perform reactions mixing samples and input reagents. Very difficult at low Reynolds number (no turbulence)

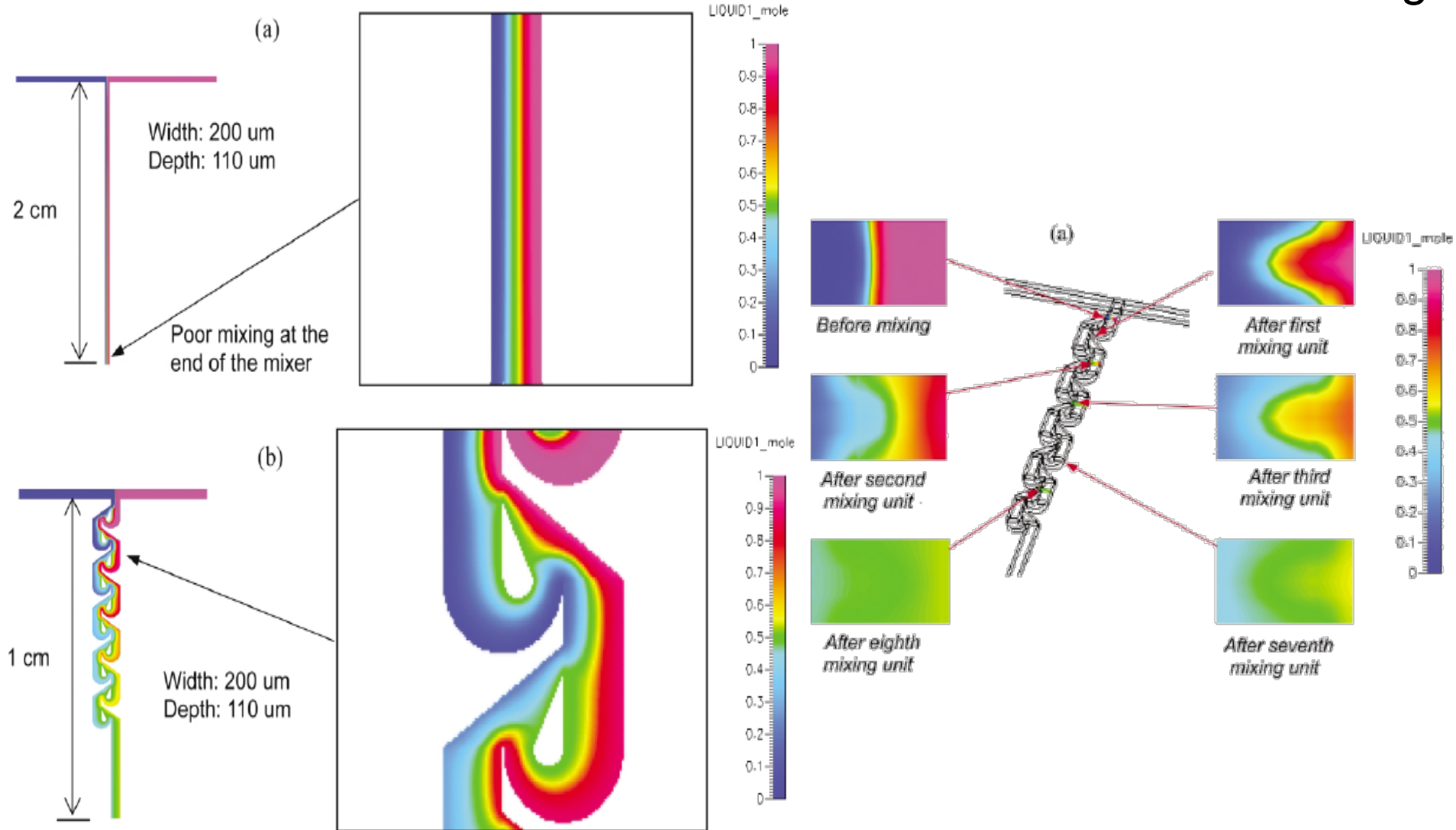
- **Passive:** *diffusion* (**slow**)
long serpentine channels
- **Active:**
 - ultrasound
 - magnetic stirring
 - bubbles





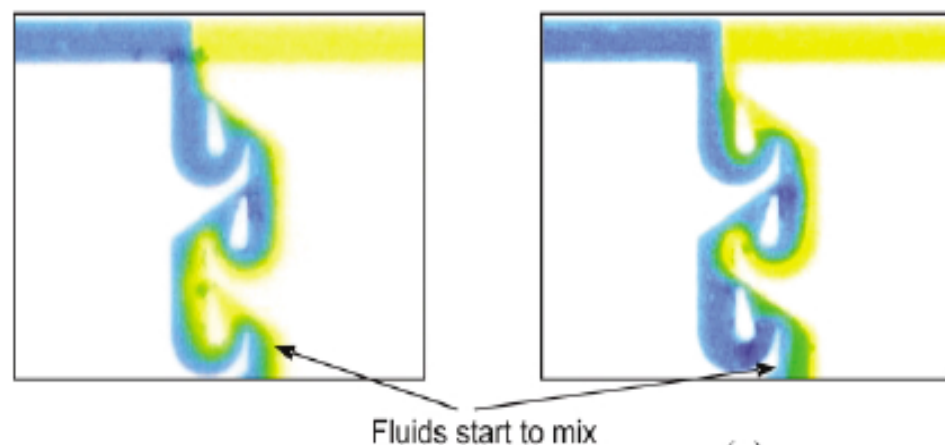
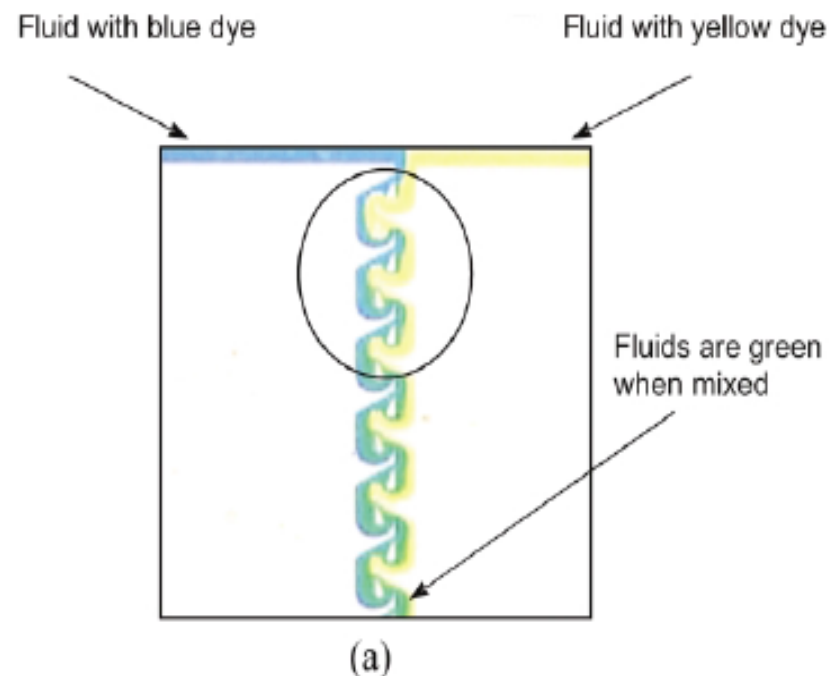
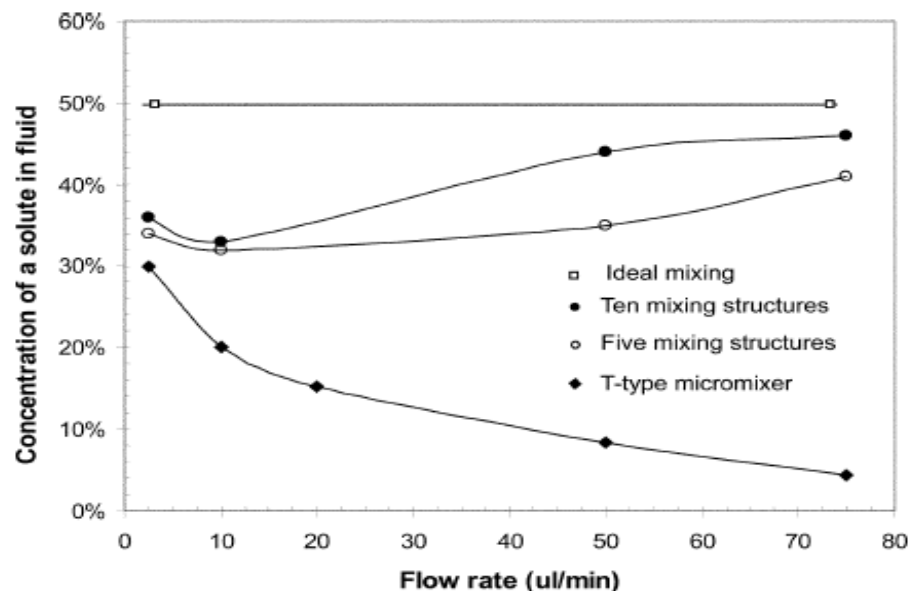
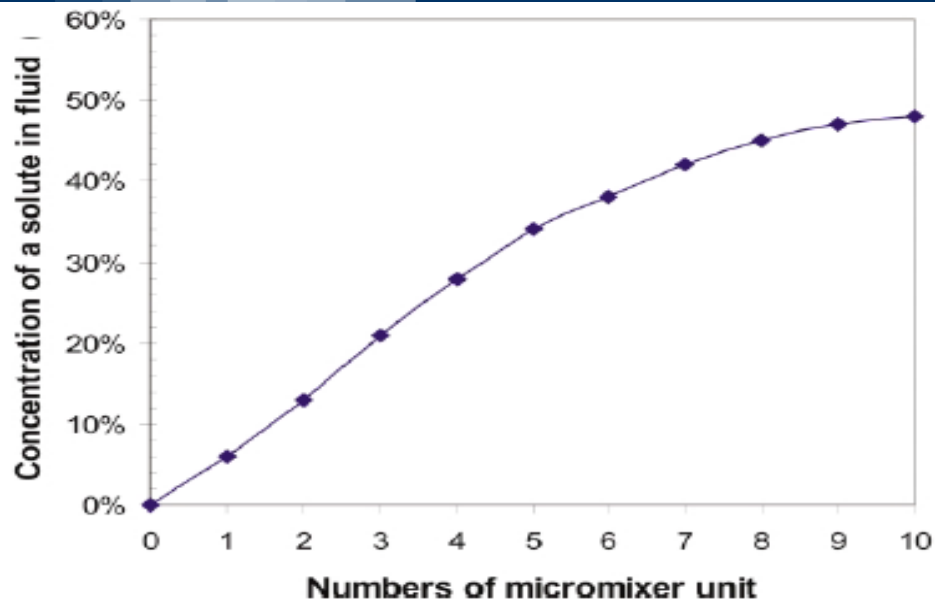
Passive Tesla Mixer

Turbulent flow due to the Tesla structure can be used for mixing





Sizing and Characterization





A Broad Family of Passive Mixers

Categories	Mixing technique	Mixing time (ms)	Mixing length (μm)	Mixing index
Lamination	Wedged shaped inlets	1	1	0.9
	Split T-shaped inlets	–	500	0.60
	Y-shaped inlets	–	13,000	0.99
<i>Chaotic advection</i>				
Intersecting channels	Unbalanced splits and collisions channels	–	5500	0.65
	Rhombic channels with flat angles	–	15,000	0.95
	Unbalanced split and recombine channels with asymmetric sub-channels	–	10,300	0.58
	Staggered curved-channels	–	77,000	0.96
Convergent–divergent channels	Split and recombination channels with convergent–divergent walls	–	6720	0.90
	Convergent–divergent sinusoidal channels	–	22,000	0.99
	Convergent–divergent channels with pulsatile flow	–	24,000	0.92
3-D structures	Interconnected multi-channel network	–	10,000	0.98
	3-D crossing manifold	–	250	0.90
	3-D periodic perturbation	–	50,000	0.99
	3-D spiral	–	2340	0.99
	H-shaped sub-channels	–	21,000	0.98
	3-D triangle	–	300	0.90
Embedded barriers	Static mixing elements (SME)	1	185	–
	Obstruction-pairs	–	4200	0.9
	Rib-shaped	–	1344	0.95
	Inserted diamond-obstacles	–	10,000	0.96
	Simple periodic geometric features	–	14,000	0.79
	Embedded-obstacles with notched walls	80	–	0.90
	Shifted trapezoidal blades	–	5000	0.95
Staggered herringbone structures (SHM)	Grooved	–	3000	0.83
	Grooved	–	5000	0.95
Twisted channels	Clamped capillaries	–	313.6	0.97
Tesla structures	Modified	–	2370	0.90
	3-D Tesla	–	11,200	0.94

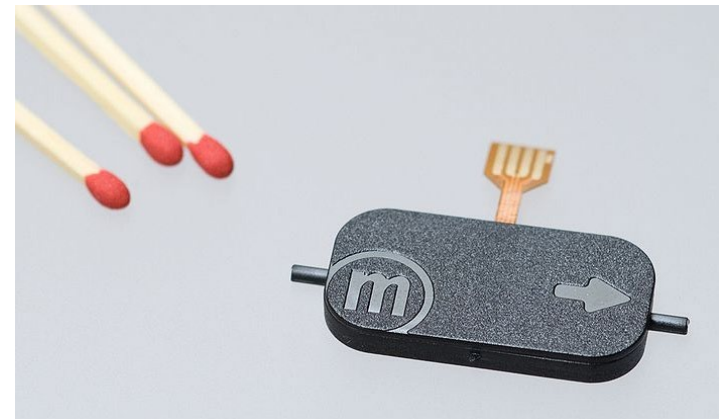


The Beating Heart of the LOC: the Pump

Generates the pressure the drives the liquid.
Controls the speed (with the flow rate)

Features:

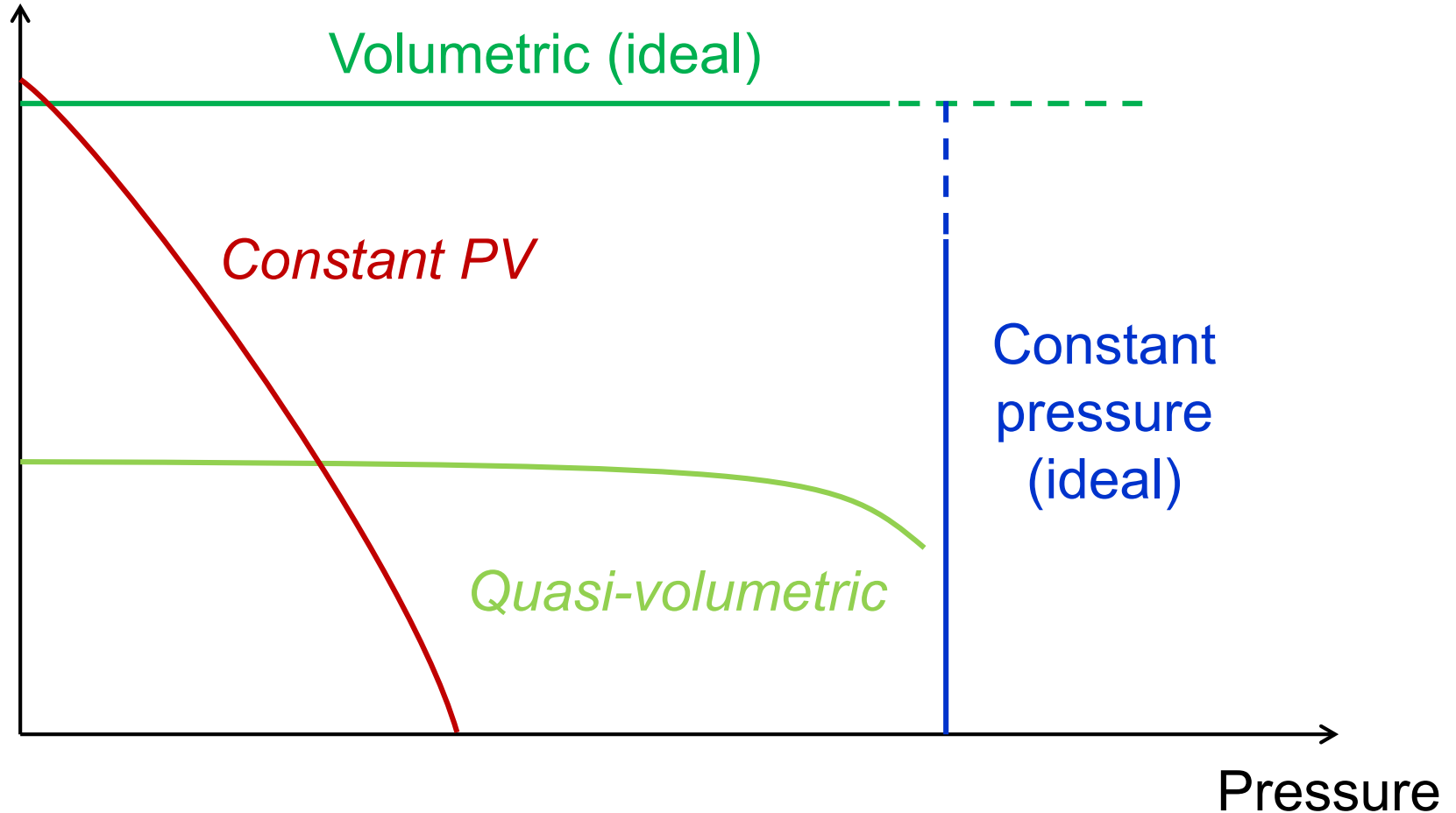
- Pressure
- Range of flow rates
- Max. volume
- Reverse flow
- External vs. embedded in the chip
- Bulkiness (size, weight)
- Power consumption





Pumps Characteristic Curve

Flow rate



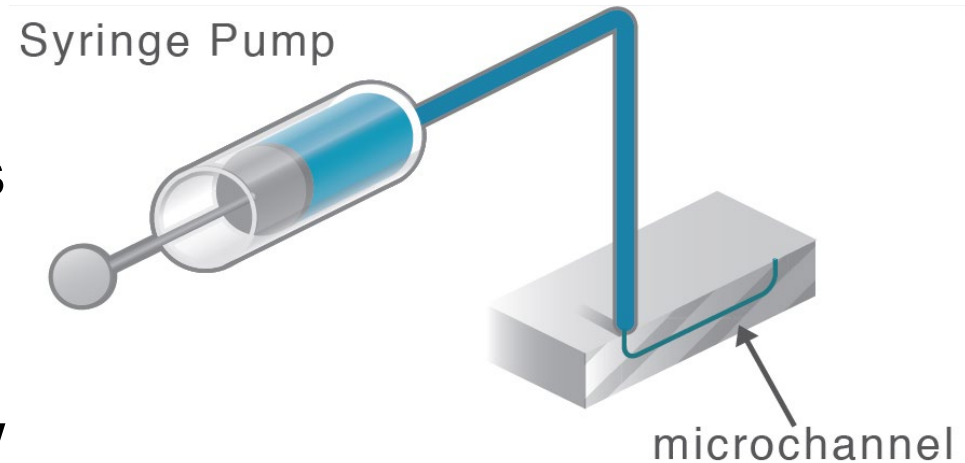
External Pumps: Syringe Pump

A stepper motor pushes/pulls the syringe plunger

Fine adjustment of the flow rate and easily programmable profiles

Discrete or «continuous» steps

Finite pumpable volume





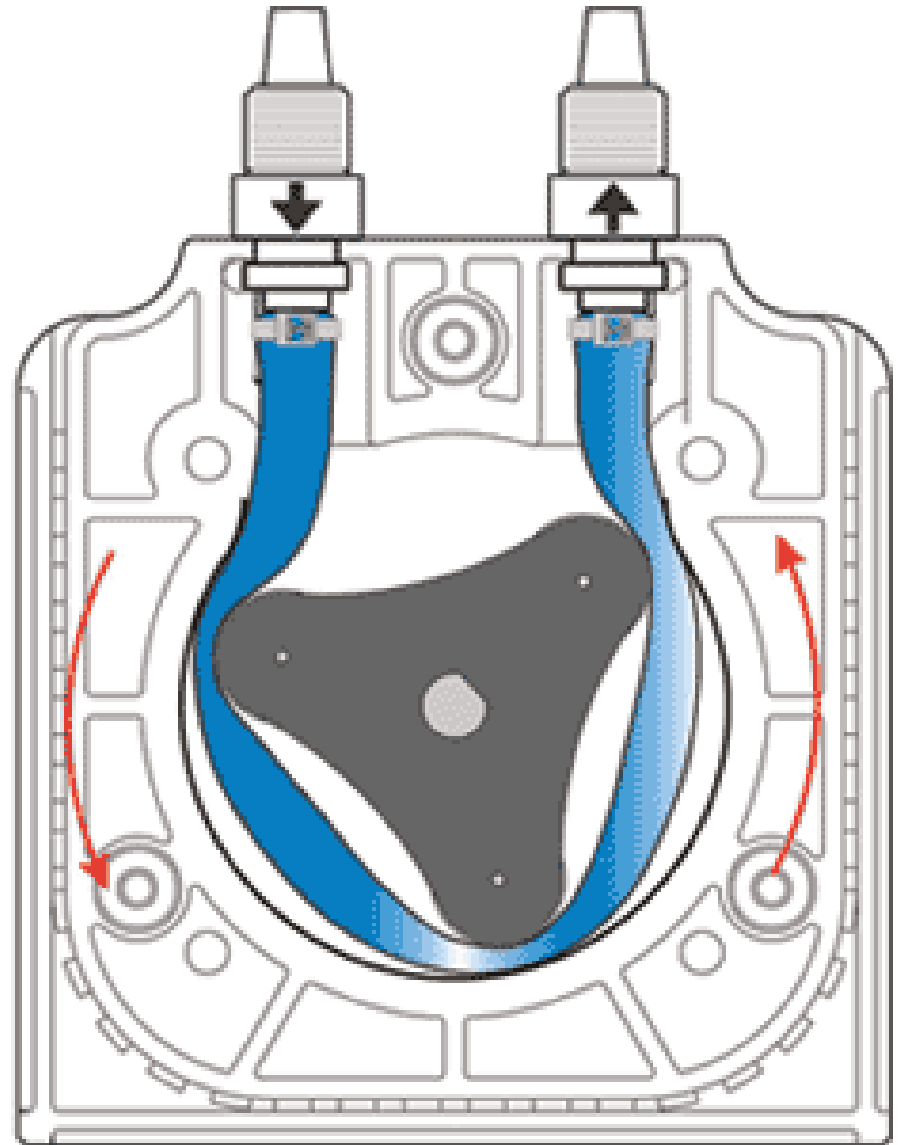
External Pumps: Peristaltic Pump

Rotary mechanism
compresses a flexible tubing

Periodic pulses

(mimic biological systems)

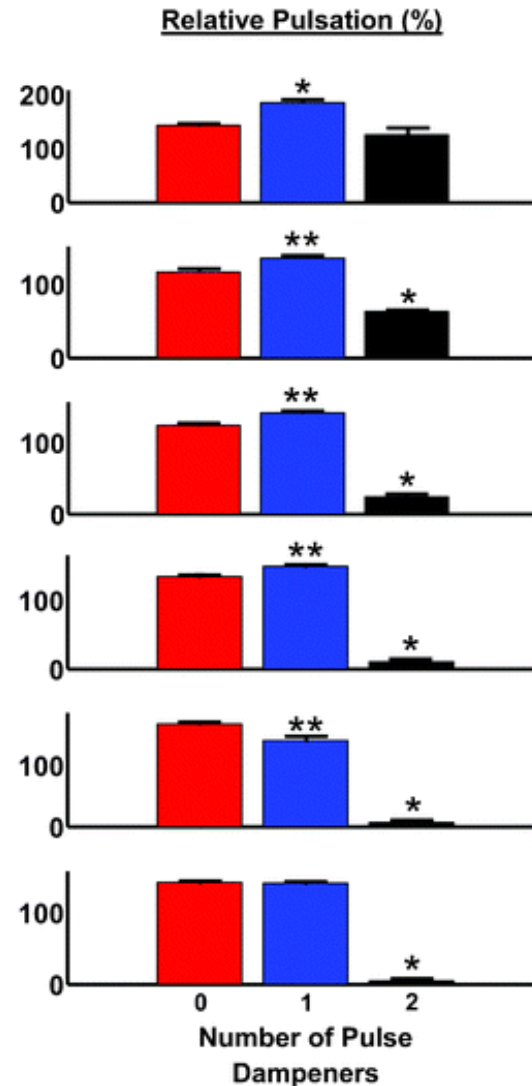
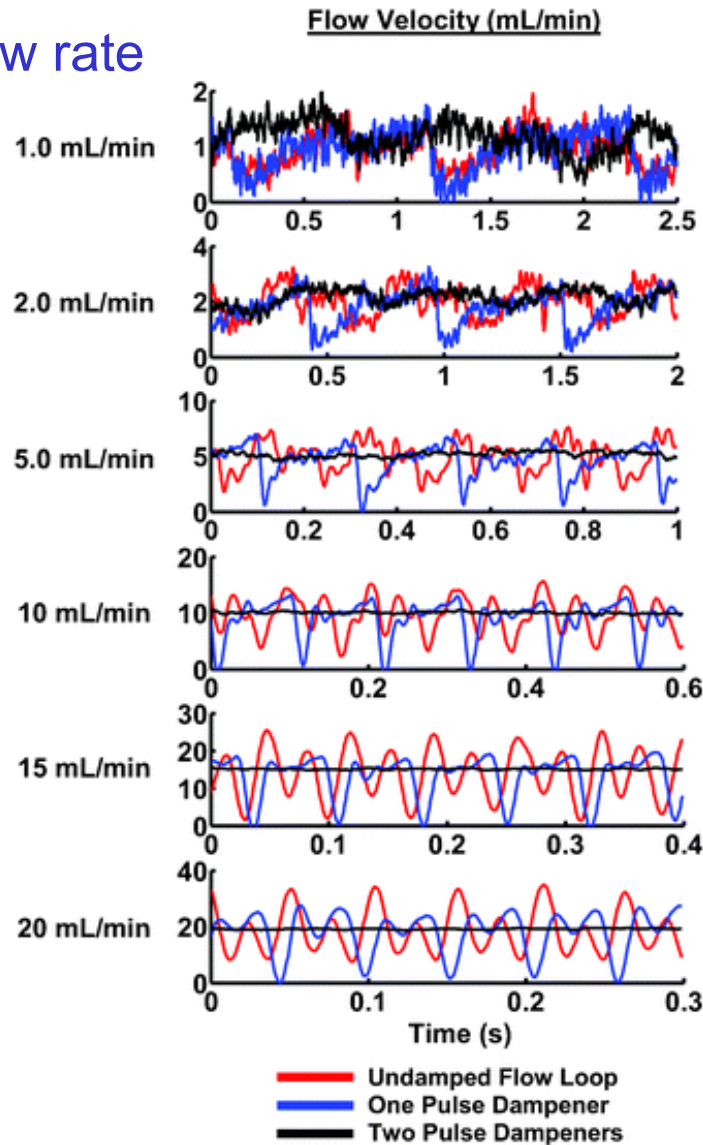
Unlimited liquid volume





Main Drawback: the Pulsatile Flow

Flow rate



Periodic
fluctuations
can be low-
pass filtered



Fluidic-LPF

Flow Stabilization via Fluidic-LPF

Preload Effect (via FCSP)

Syringe Pump

Targeted Microfluidic Device

Stabilization

Q_{in}

Q_{out}

$Q_{out}(t)$

$P_A(t)$

C_{ACU}

C_{Tube}

(a)

(b)

Air

Liquid

GND

Qout (mL/h)

0

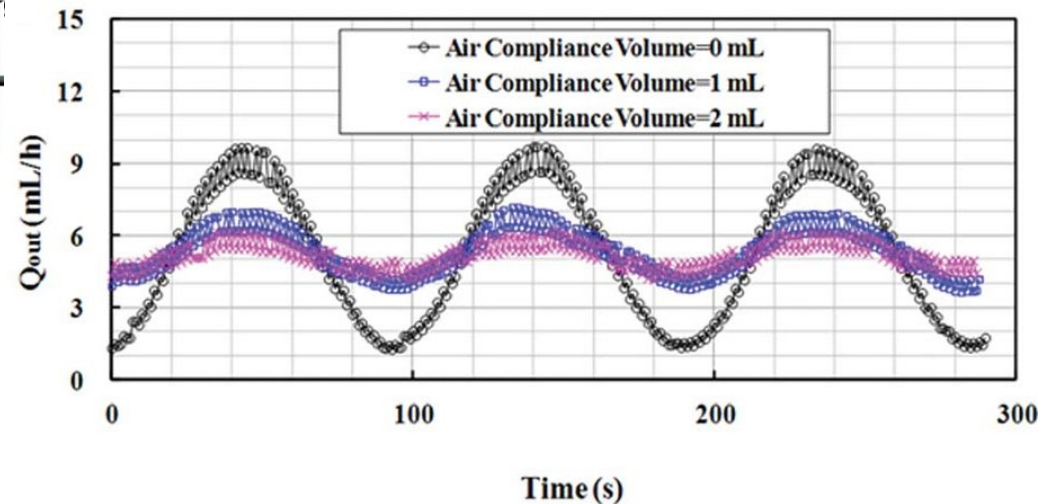
3

6

9

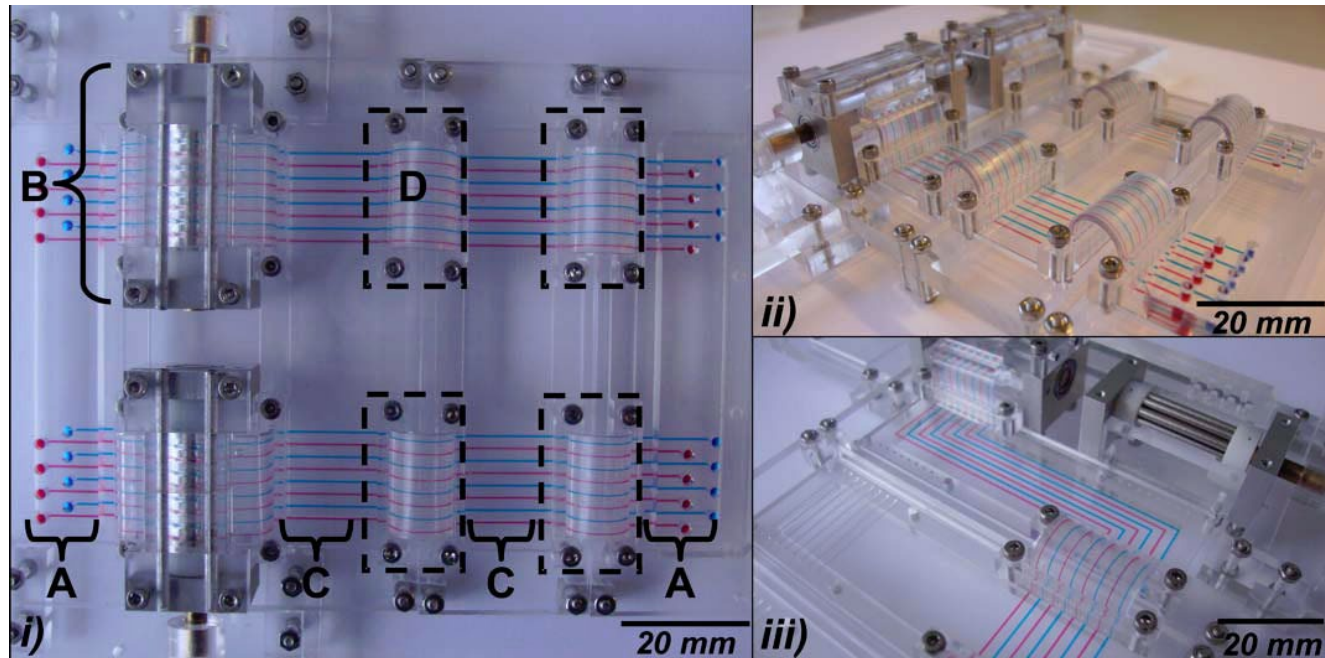
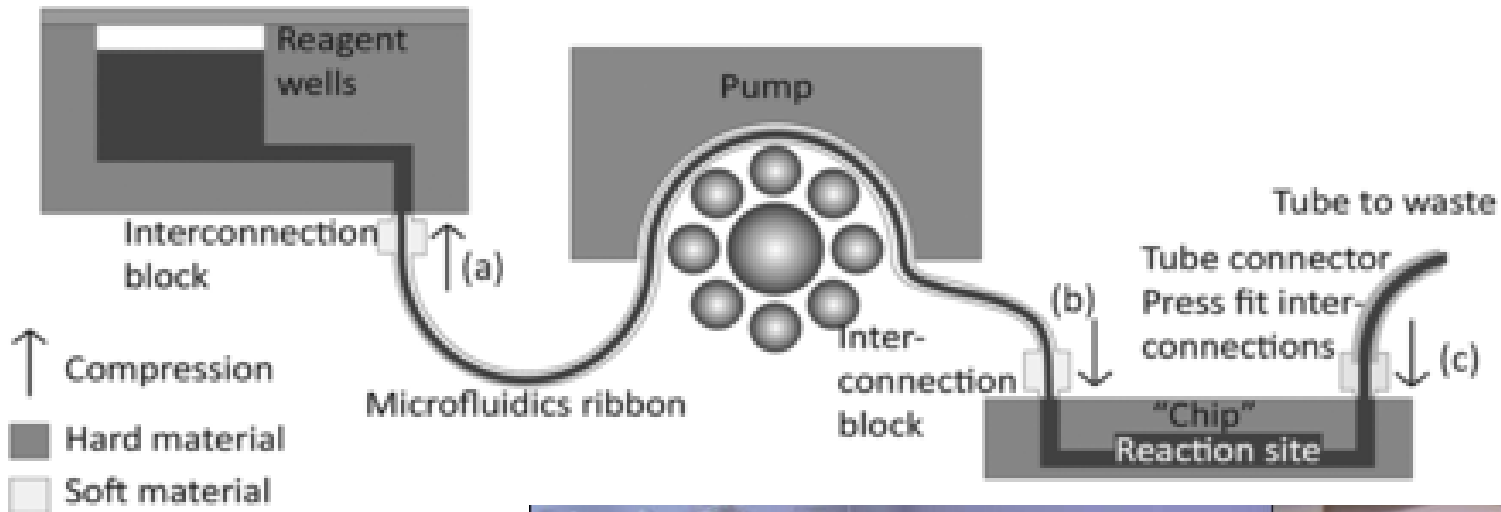
12

15



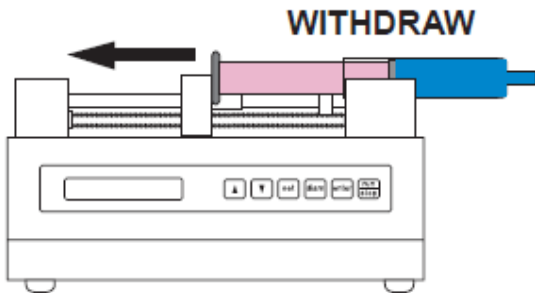
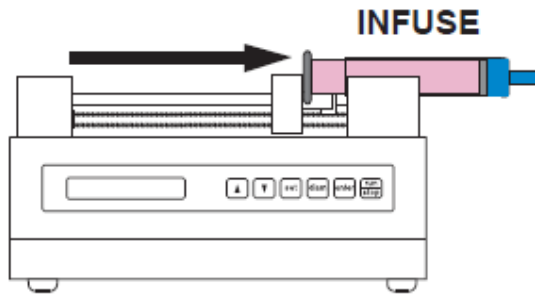


Flexible Tubings for Pumps and Connectors

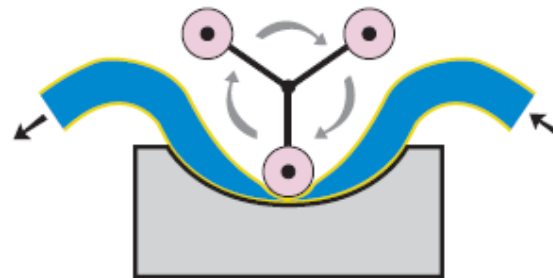
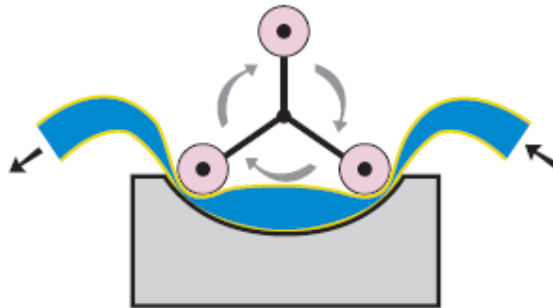


Comparison of External Pumps

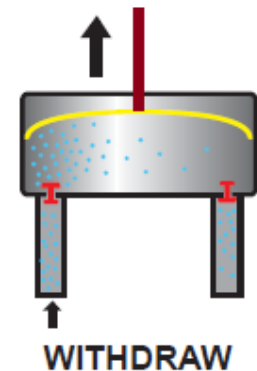
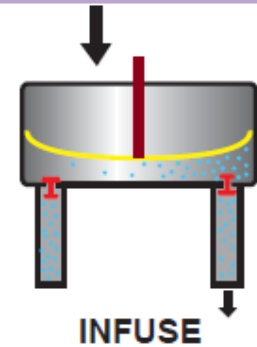
SYRINGE PUMPS



PERISTALTIC PUMPS



DIAPHRAGM PUMPS



Comparison of External Pumps

SYRINGE PUMPS

Pro:

- High pressure
- Bidirectional
- Precise control
- Multiple channels

Con:

- Finite volume

PERISTALTIC PUMPS

Pro:

- Continuous volume
- Bidirectional
- Sterilizable
- Simple/low cost

Con:

- Pulsatile flow
- Low pressure
- Moderate precision

DIAPHRAGM PUMPS

Pro:

- Continuous volume
- Simple
- Chemical resistant
- Low maintenance

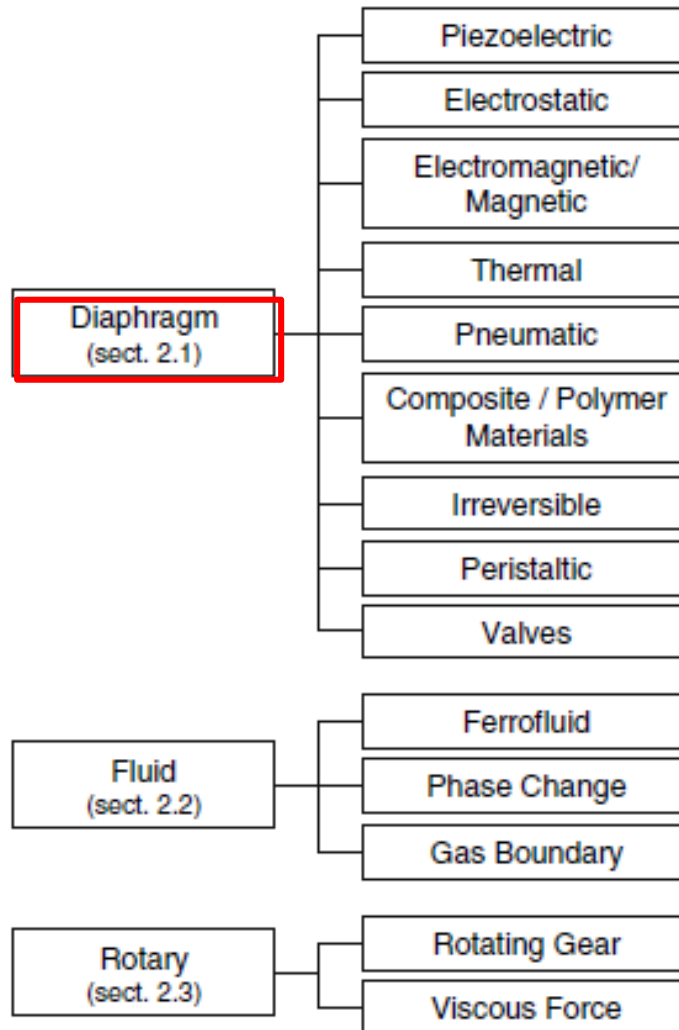
Con:

- Pulsatile flow
- Monodirectional
- Low pressure

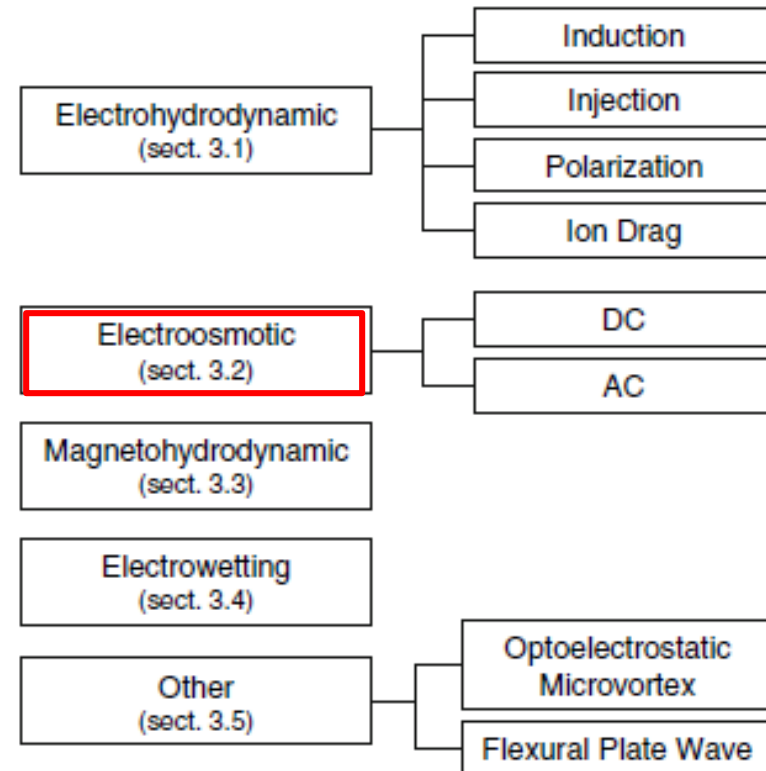


Integrated Micropump Technologies: 2 Families

Mechanical Displacement
(Moving boundary or liquid displacement pressure force, sect. 2)

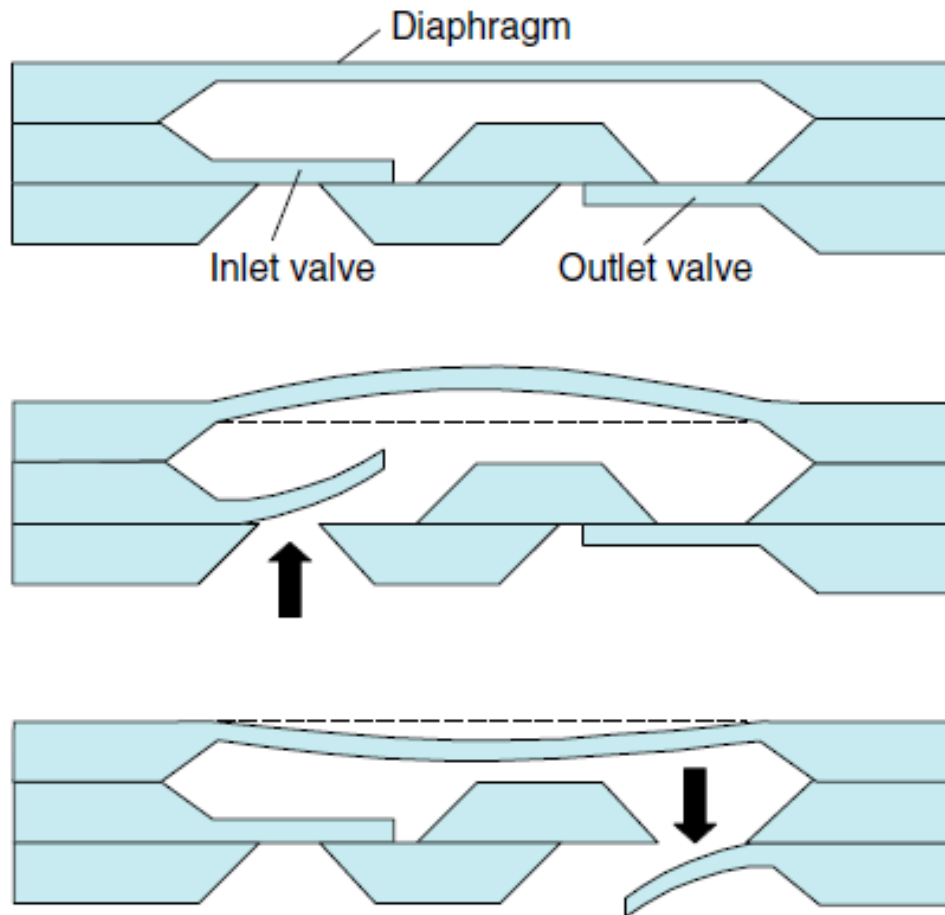


Electro- and Magneto-kinetic
(Continuous, direct energy transfer, sect. 3)





Diaphragm Displacement Micropumps



Mechanical movement of a solid liquid boundary (membrane, like a valve) combined with 2 valves for fluid rectification

Expansion stroke:
lower pressure

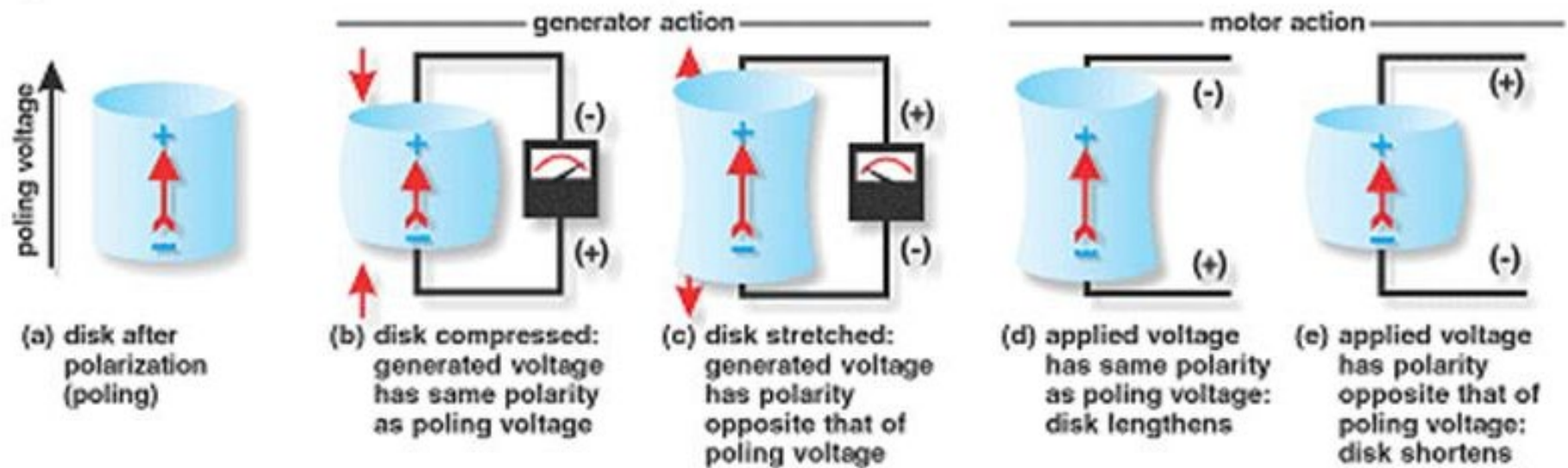
Compression stroke:
higher pressure



Piezoelectricity

In piezoelectric crystals (quartz, ceramics):

mechanical stress (force) ↔ Reversible ↔ charge (electric field)

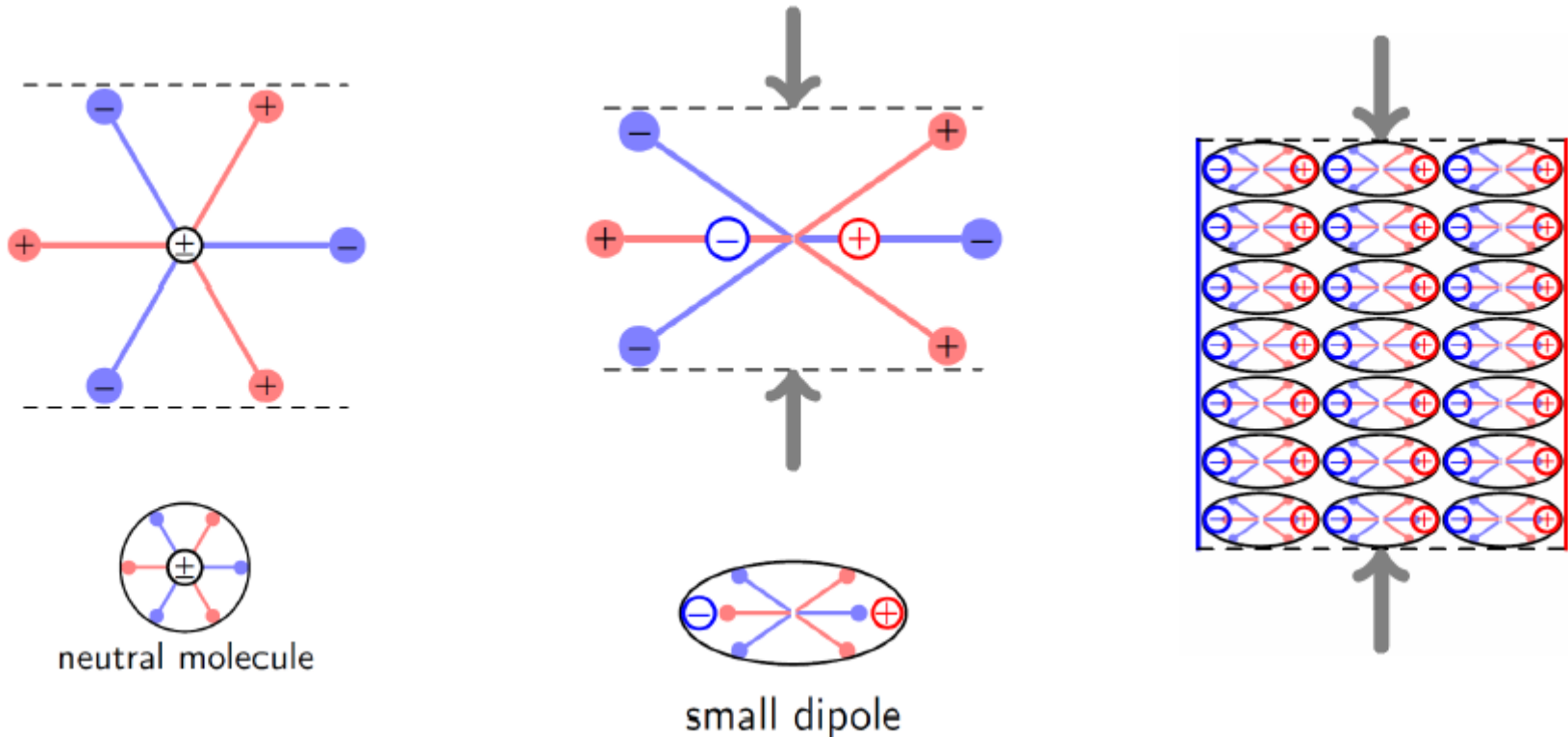


Example: slab of PZT (lead zirconate titanate) 1cm x 1cm x 1mm
z: 370pC/N, x: 110pC/N,
1V Applied $\rightarrow \Delta z = 0.37\text{nm}$ (very precise motion control)



A Molecular Model

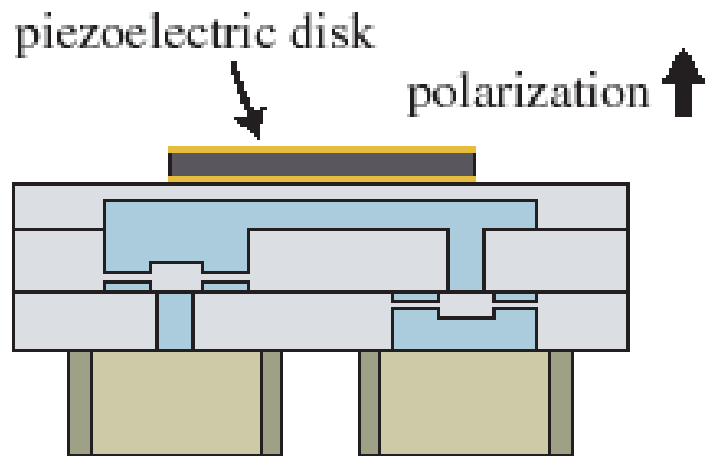
Only for crystals **without symmetry center**: mechanical force deforms distribution of charges, creating dipoles



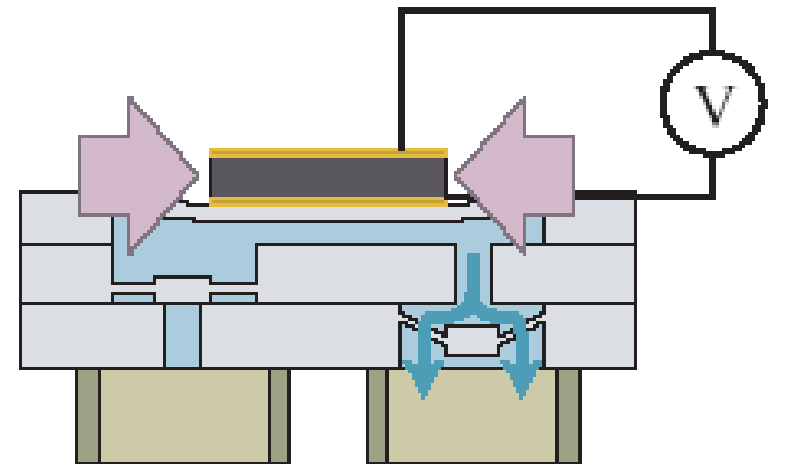
Piezo materials offer very good mechanical properties, but require high pressures and temperatures, difficult to integrate in microfabrication processes.



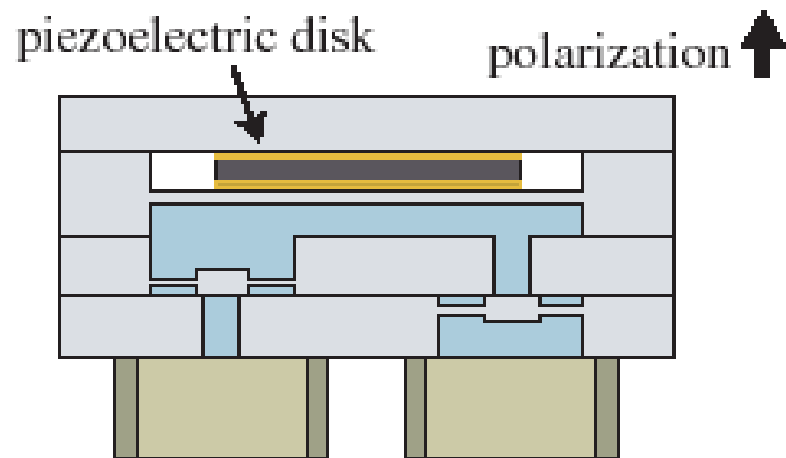
Piezo Actuation



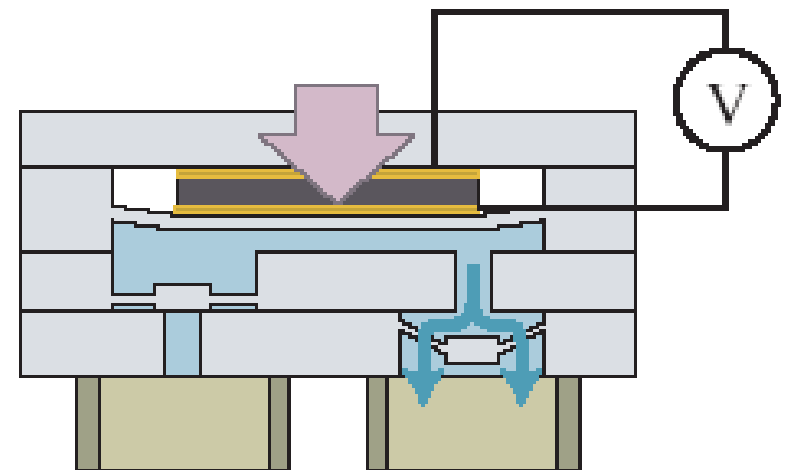
(a)



discharge stroke



(b)

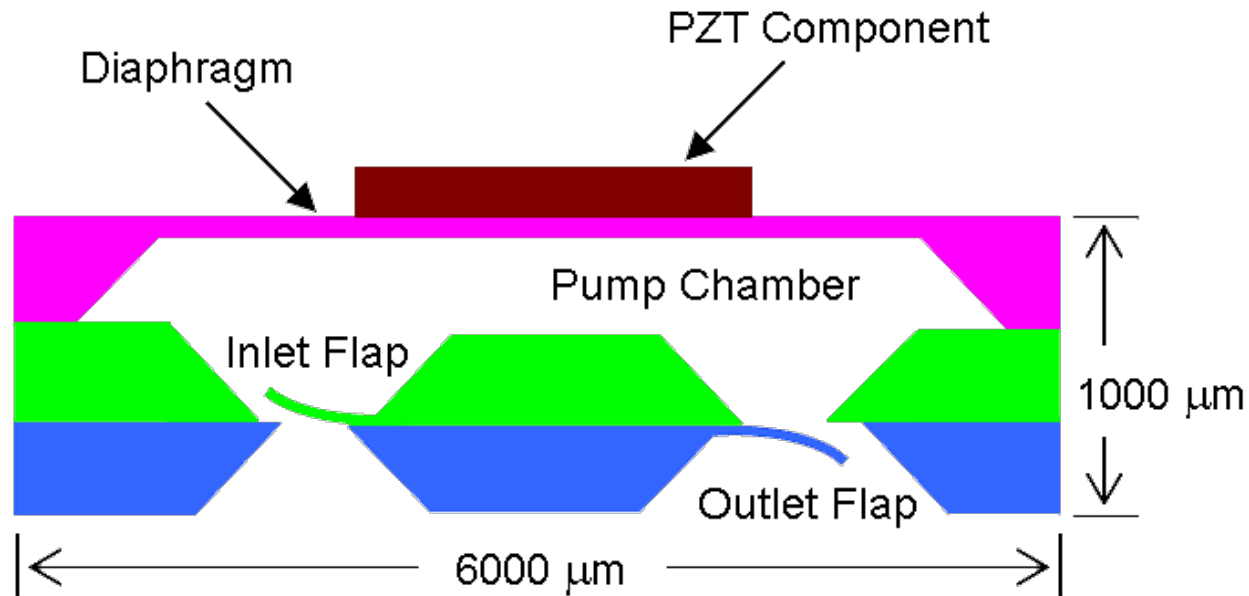


discharge stroke



Membrane Piezo Integrated

Size:



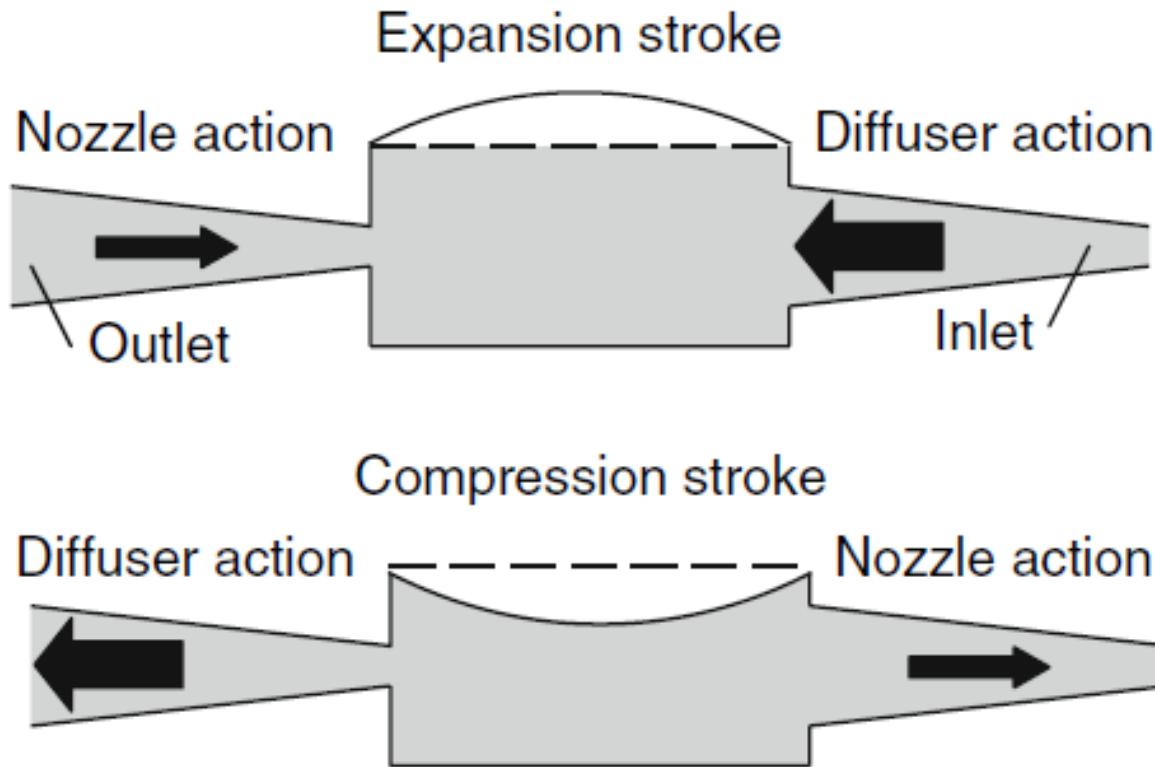
Performances:

- 1 stage: $1\ \mu\text{l}/\text{min}$ pumped with 50V applied across $63\ \mu\text{m}$
- 4 cascade stages at 14kHz, $3\ \text{ml}/\text{min}$, pressure $7\ \text{kPa}$



Fixed Geometry

No moving parts for the fluid rectification, asymmetry of a conic duct (nozzle/diffuser action)

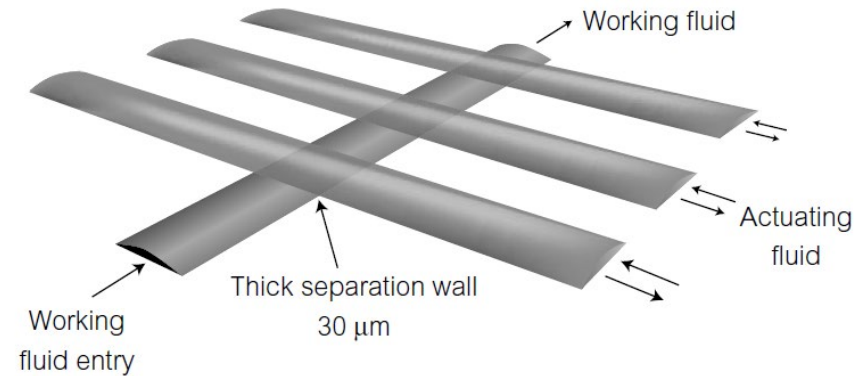


Monodirectional

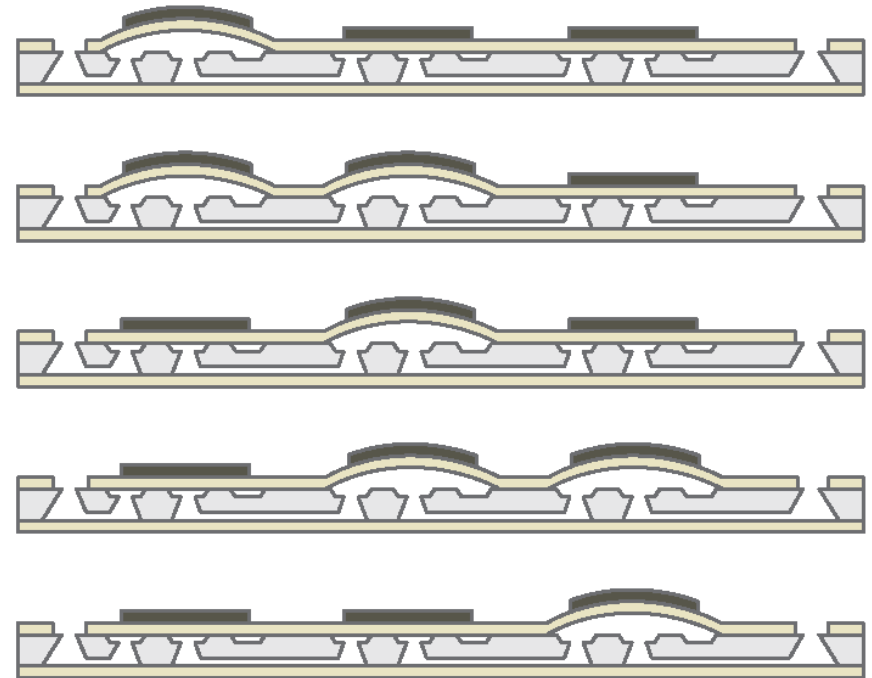


Peristaltic Micropumps

A cascade of 3 membrane microvalves, properly driven in series



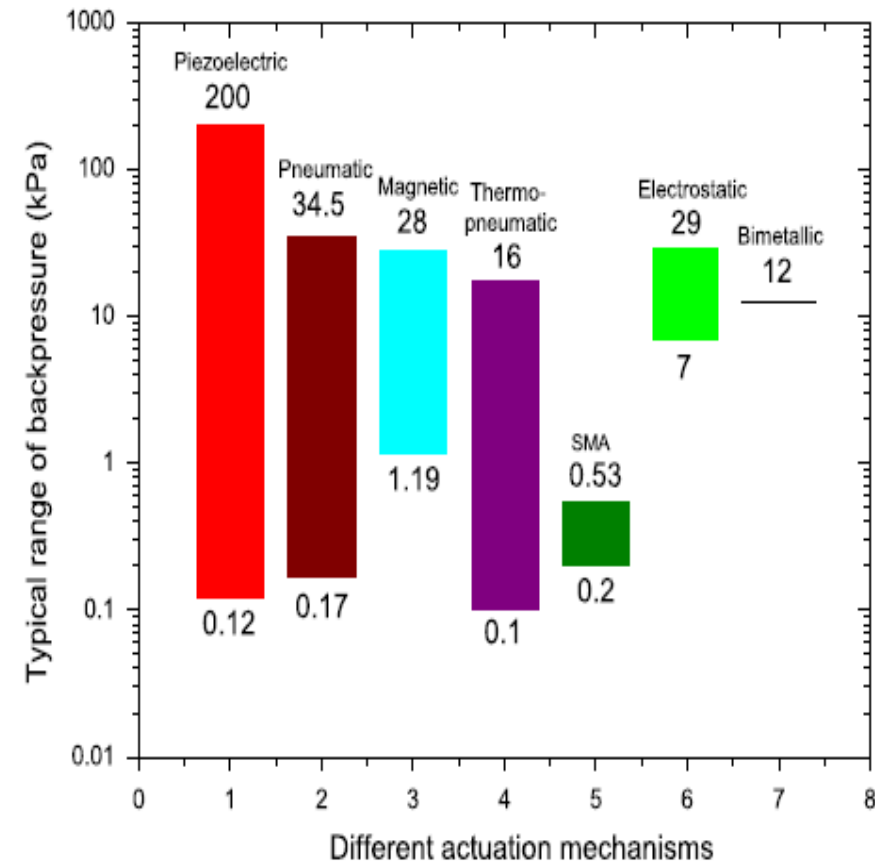
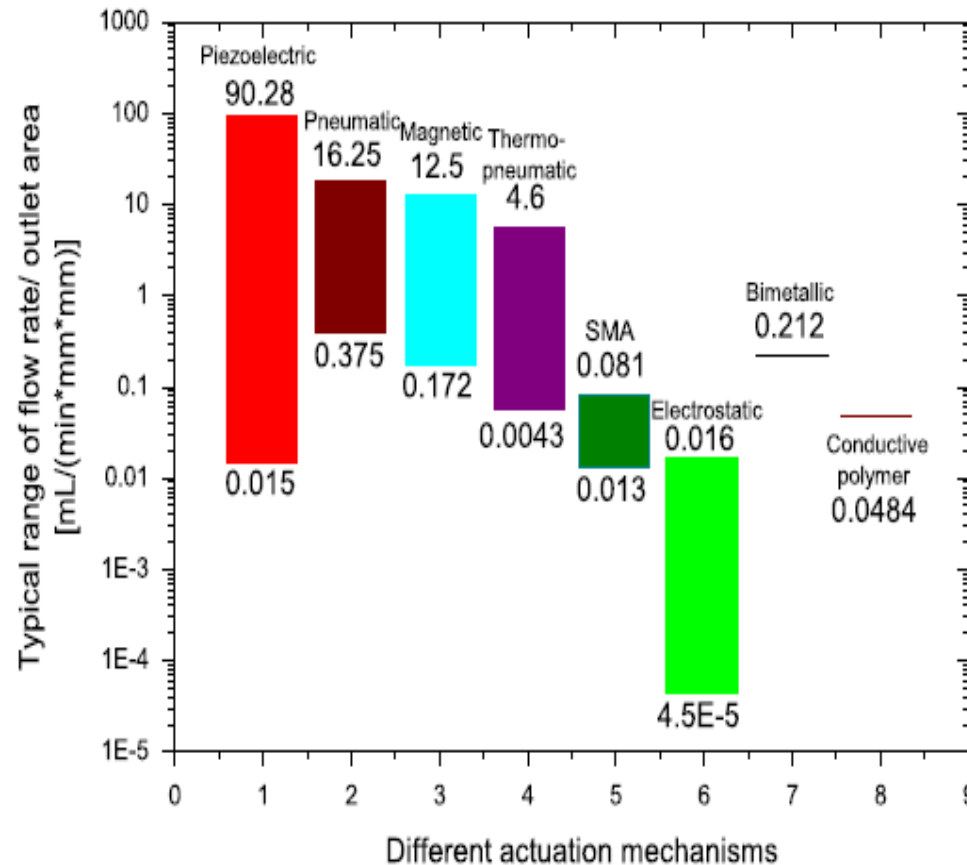
Activation sequence:





Micropump Technology Selection Criteria

Achievable **flow rate** and **pressure** ranges:



Piezoelectric is the most common choice thanks to a wide range



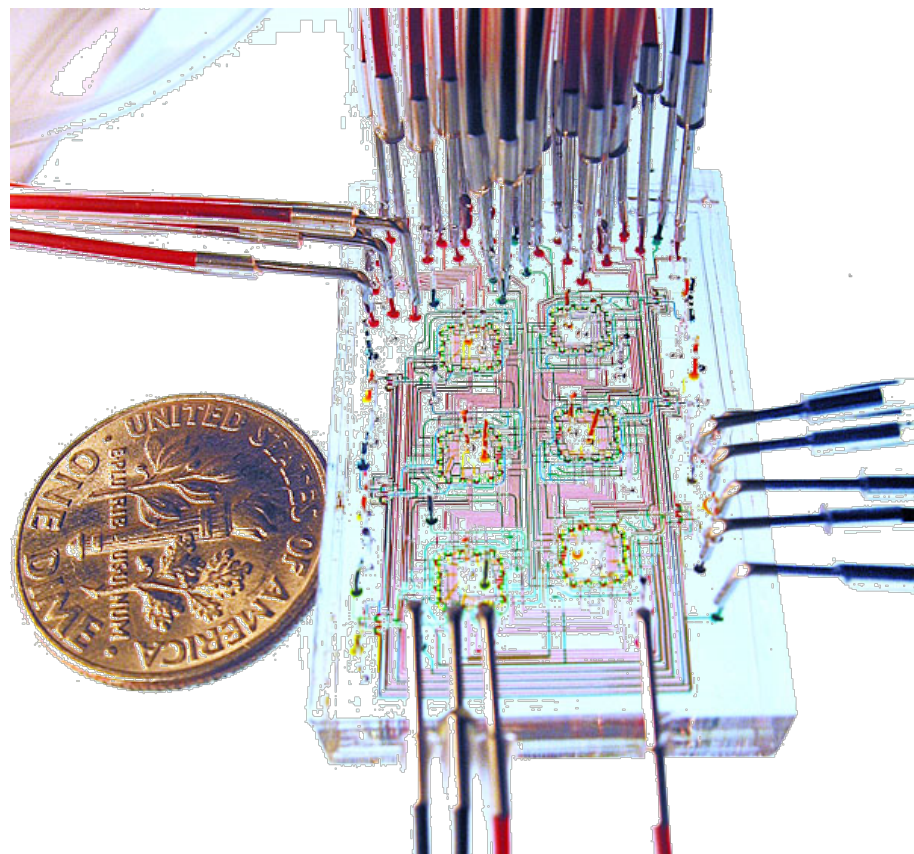
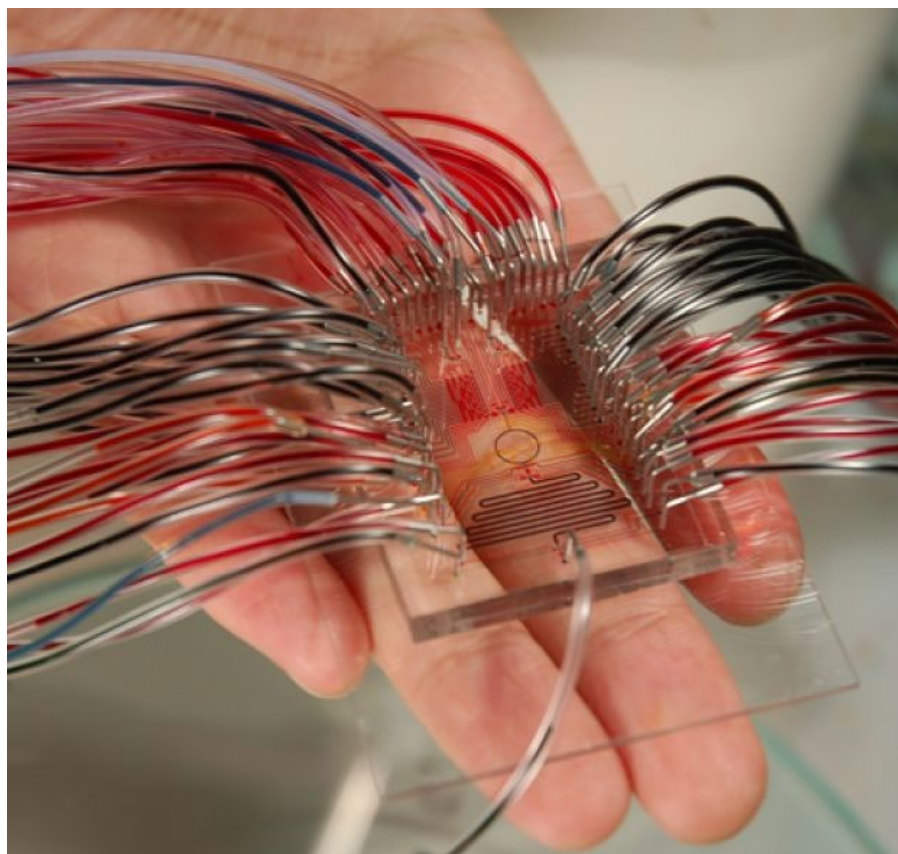
Selection Criteria: Driving Signals

Actuation schemes	Energy density (J/m ³)	Efficiency	Frequency (Hz)	Voltage (V)	Power (mW)
SMA	2.57×10^7	0.01	<100	2–5	180–630
Conductive polymer	3.4×10^6	0.6	<1,000	1.5	180
Electrostatic	1.8×10^5	0.5	<10,000	50–200	N/A
Electromagnetic	4.0×10^5	<0.01	<1,000	3–14	13–7,000
Piezoelectric	1.2×10^5	0.3	<5,000	20–1,200	3–400
Bimetallic	4.0×10^5	10^{-4}	<100	16	N/A
Thermopneumatic	5.0×10^5	0.1	<100	6–20	~2,500



Interfacing with the Rest of the World

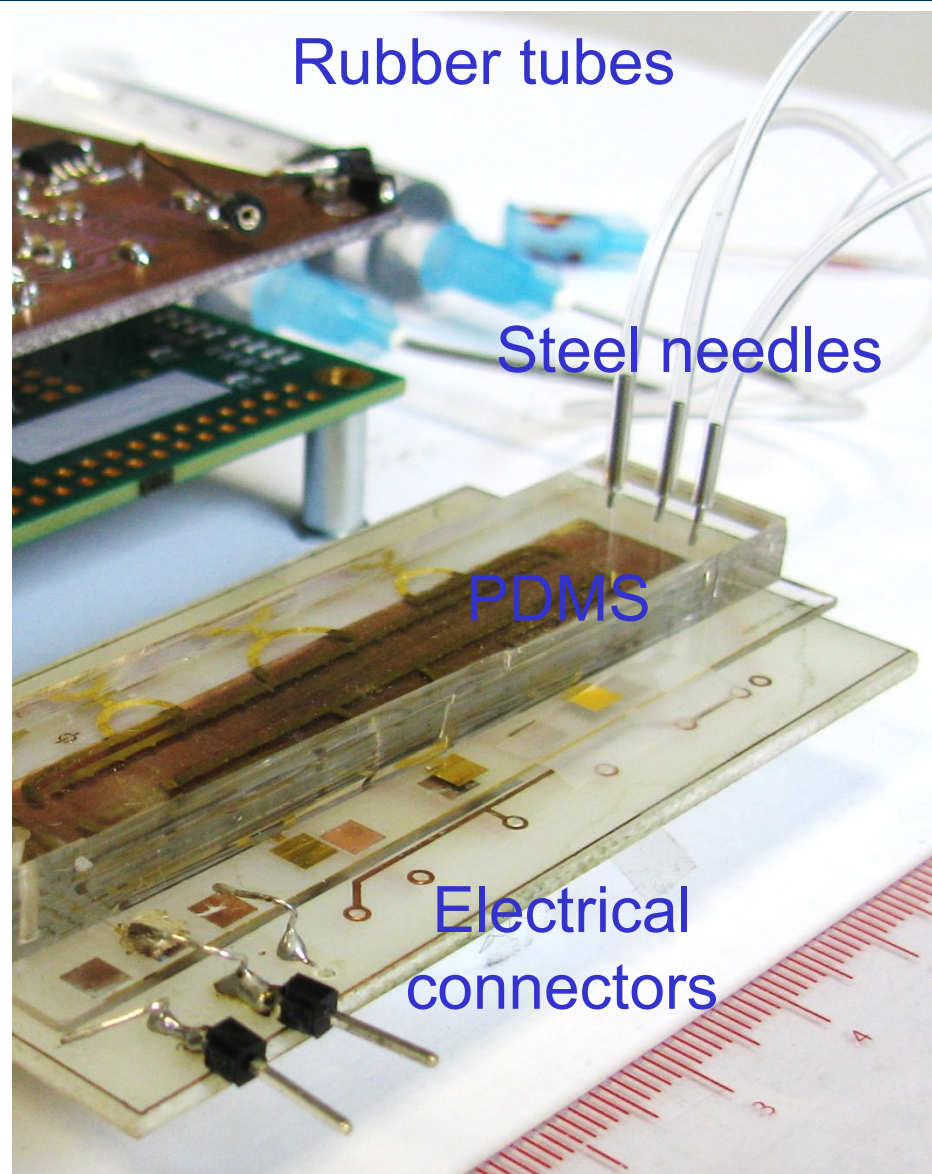
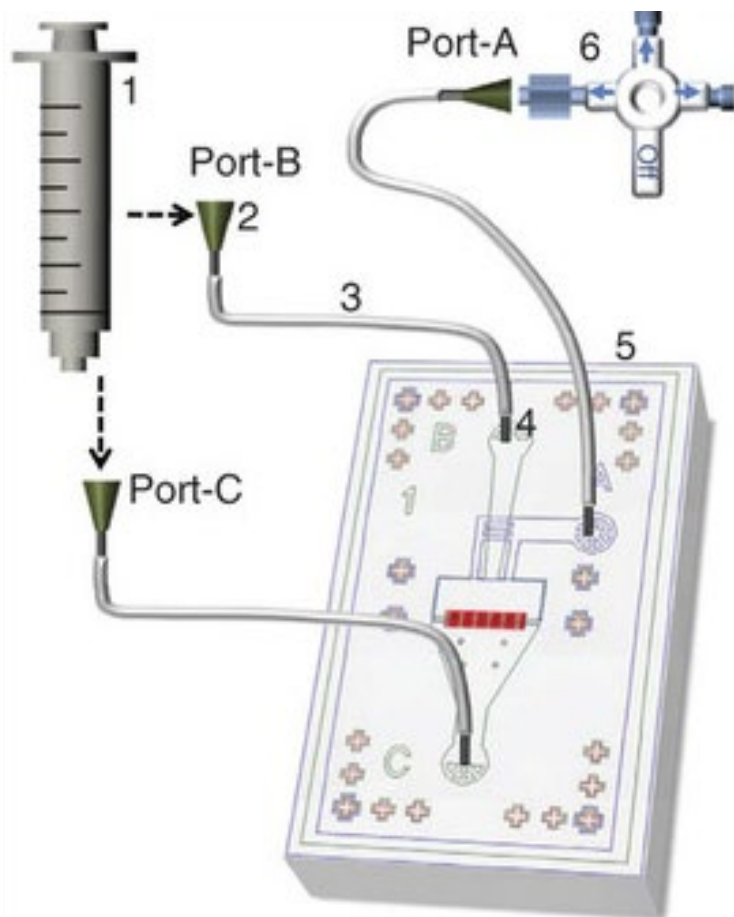
Fluidic connectors represent an open problem: several custom solutions are proposed, but an industrial standard is missing





Tubing Connections

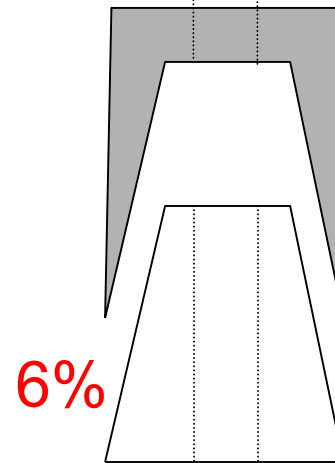
For *elastomers*: watertight with pressure



Luer Taper: a Simple Standard

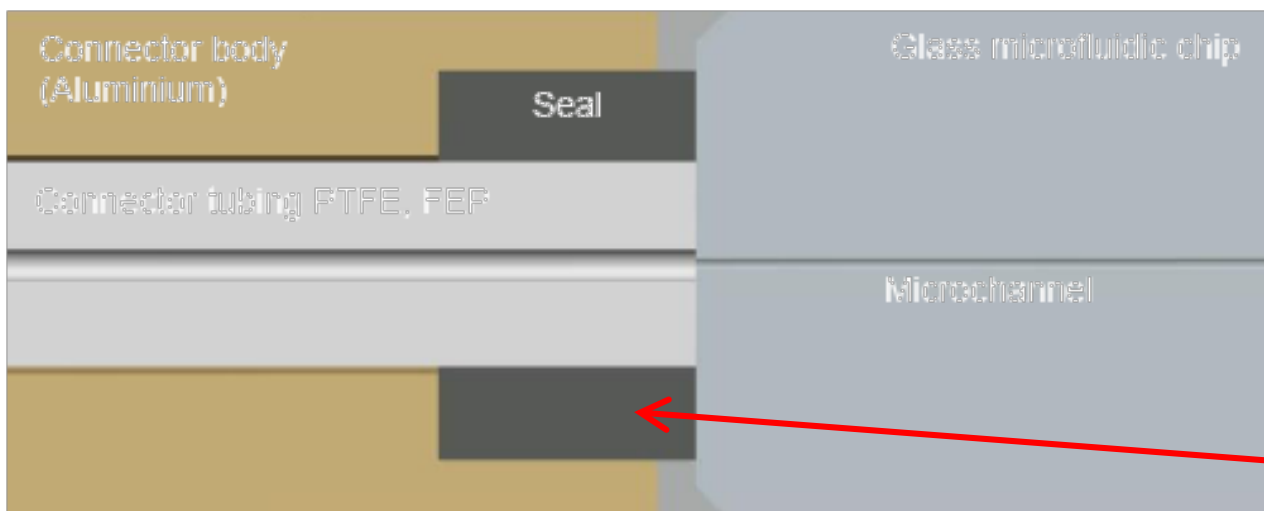
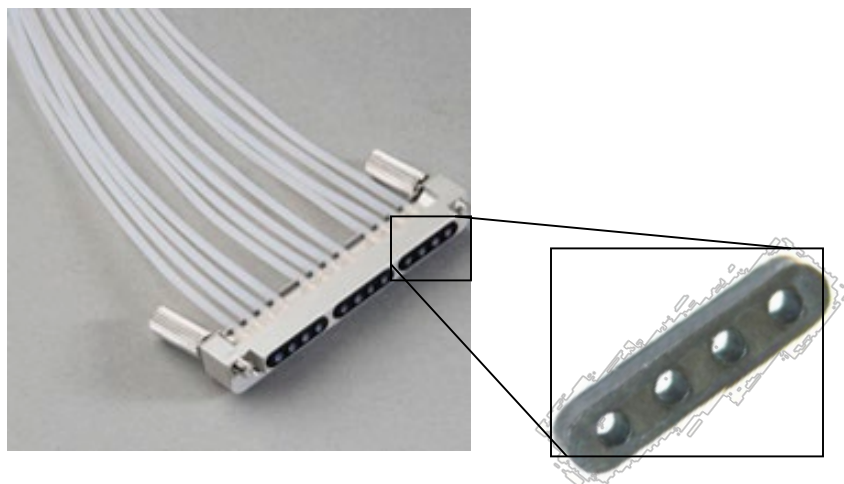
For *rigid plastic* (mainly syringes):

- 6% conical taper
- Luer lock is watertight





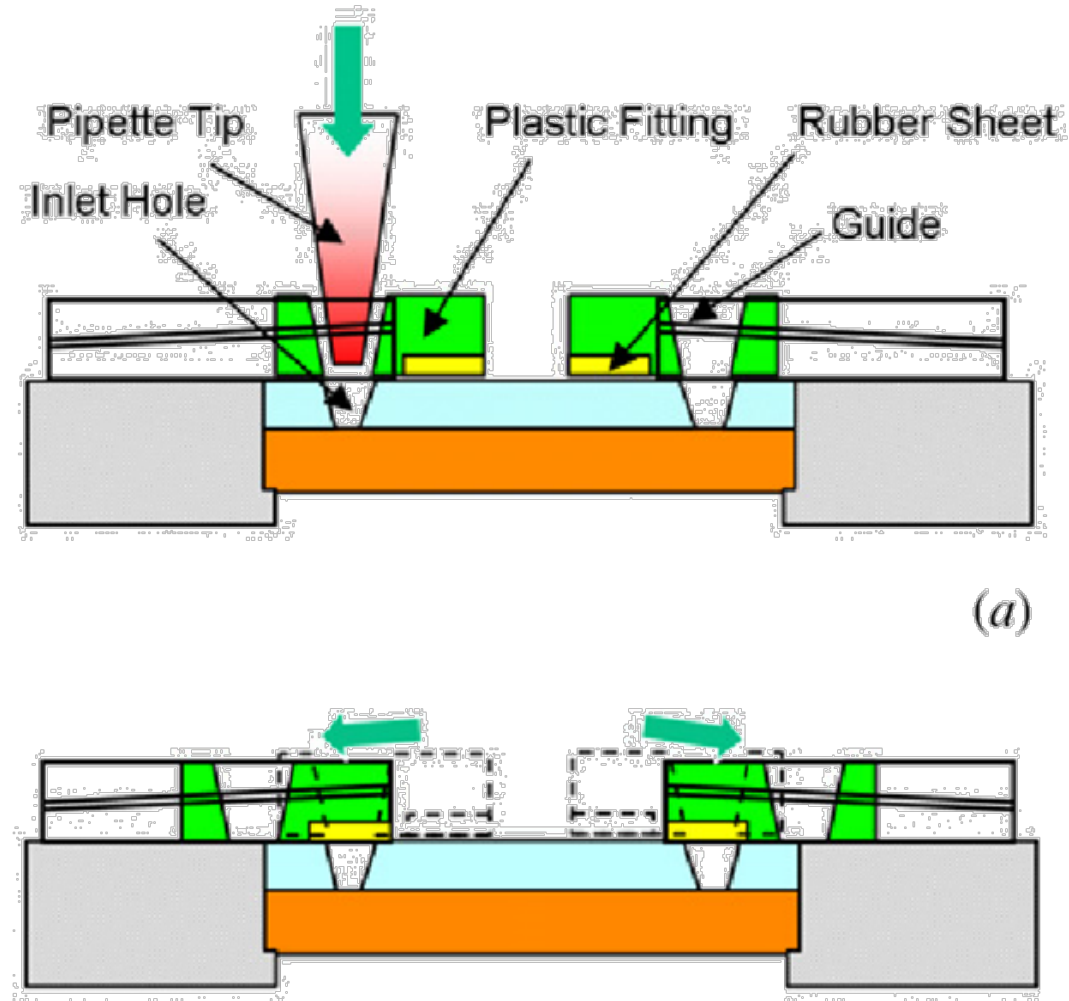
Multiple Connectors



• O-ring sealing

Sample Loading: the Pipette

- Precise volume control (1-10/10-100/100-1000 μ liters)
- Disposable tips
- Avoid bubbles





Key Take-Home Messages

- Basic microfluidic components are valves (switches), mixers, and pumps (generators).
- Passive mixing based on diffusion is not efficient, chaotic structures (Tesla) or active solutions should be adopted.
- Active actuation on flexible membranes (for valves and pumps) can be achieved with electrostatic, pneumatic, piezoelectric, magnetic and thermal power.
- Peristaltic pumps produce a pulsatile flow (can be filtered), but there is no limit on the max injectable volume.
- Among embedded micropumps, piezoelectric technology is the most versatile.
- Waterproof interface with external tubings is critical, no standard connectors exist.



A novel in-plane passive microfluidic mixer with modified Tesla structures

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Fluidic low pass filter for hydrodynamic flow stabilization in microfluidic environments†

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REVIEW

Recent advances in microscale pumping technologies: a review and evaluation

Brian D. Iverson · Suresh V. Garimella

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REVIEW ARTICLE

Current micropump technologies and their biomedical applications

Farid Amirouche · Yu Zhou · Tom Johnson